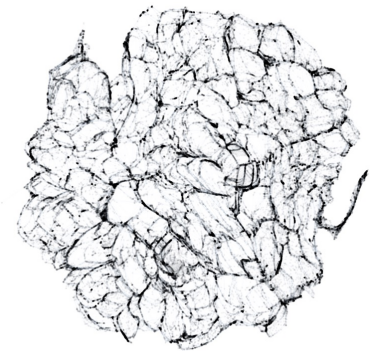
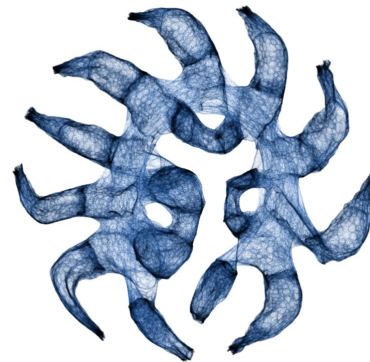
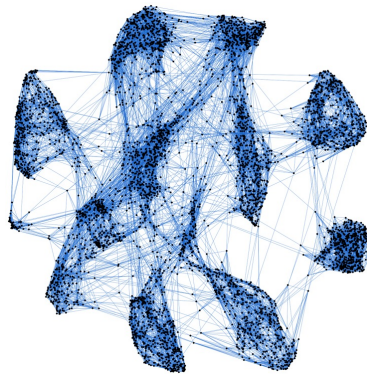
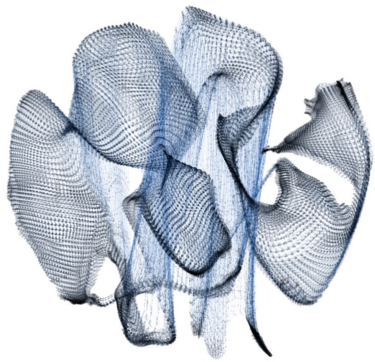


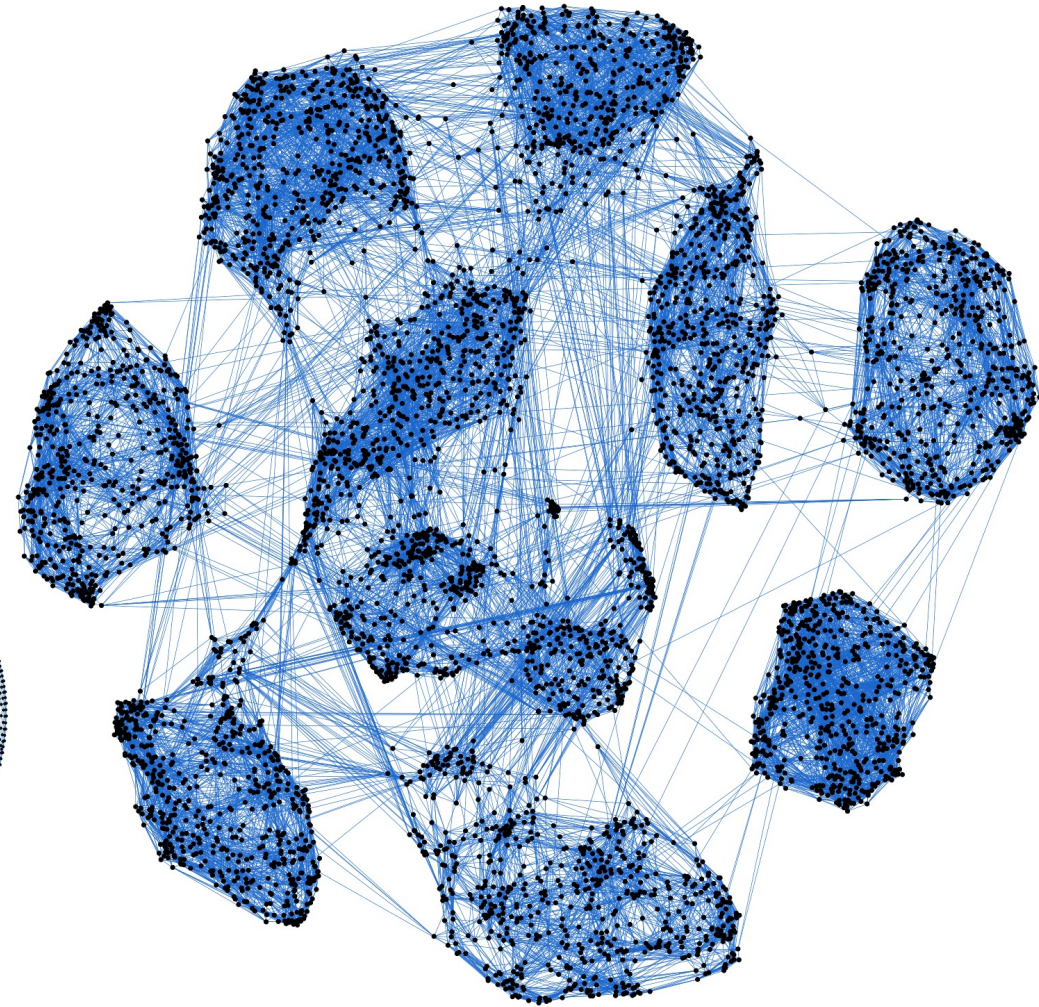
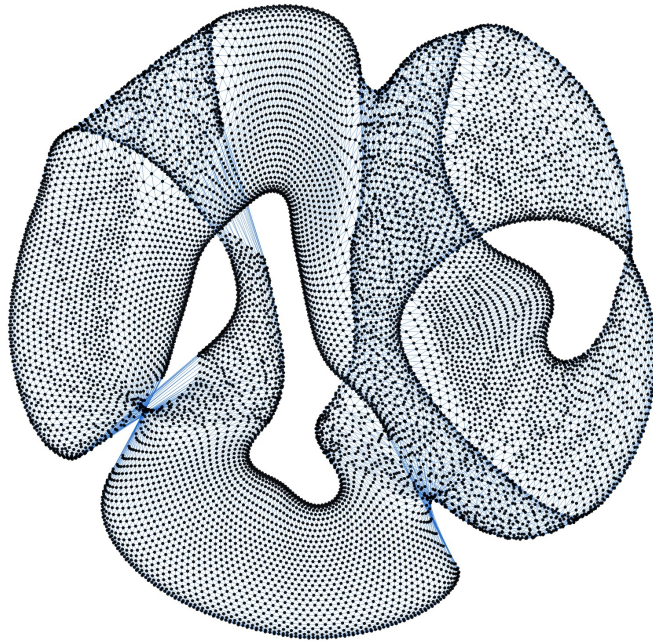
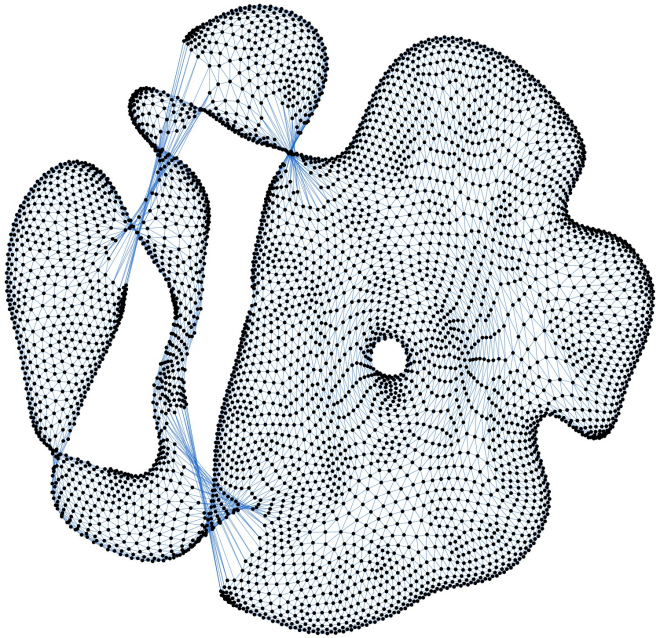
NNP-NET:

Accelerating t-SNE Graph Drawing for Very Large Graphs by Neural Networks

I. Hartskeerl, T. Mchedlidze, S. van Wageningen, P. Vangorp, A. Telea

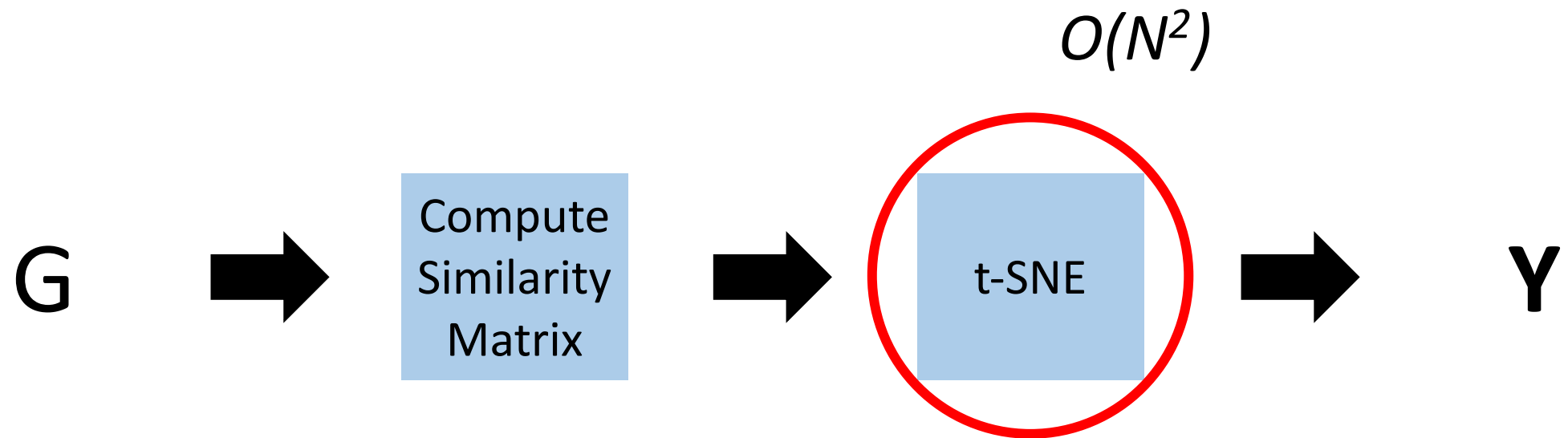


tsNET



Johannes F. Kruiger, Paulo E. Rauber, Rafael M. Martins, Andreas Kerren, Stephen Kobourov, and Alexandru C. Telea. **Graph layouts by t-SNE**. *Computer Graphics Forum*, 36(3):283–294, 2017.

tsNET



Goal:

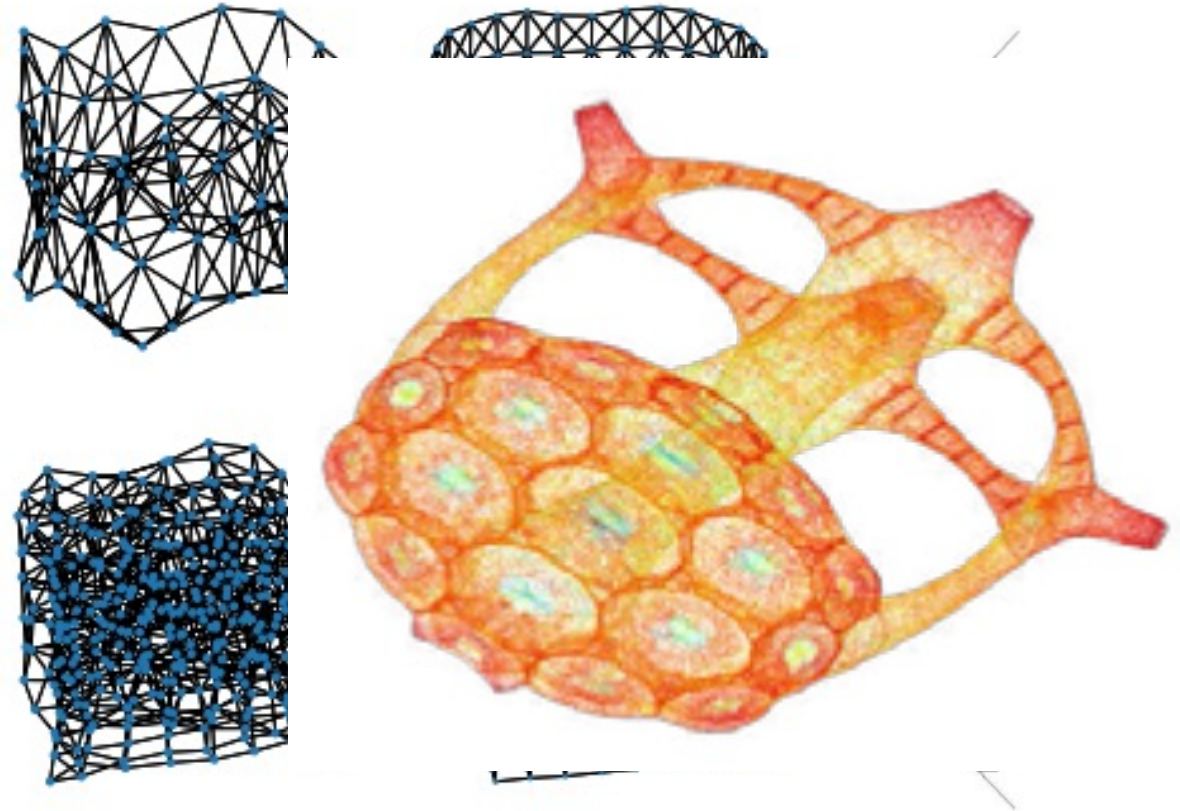
*Improve tsNET performance
By removing t-SNE from tsNET*

Aims

- Make tsNET scalable to very large graphs
 - Linear time and space complexity
- Support for weights
- Similar quality to tsNET

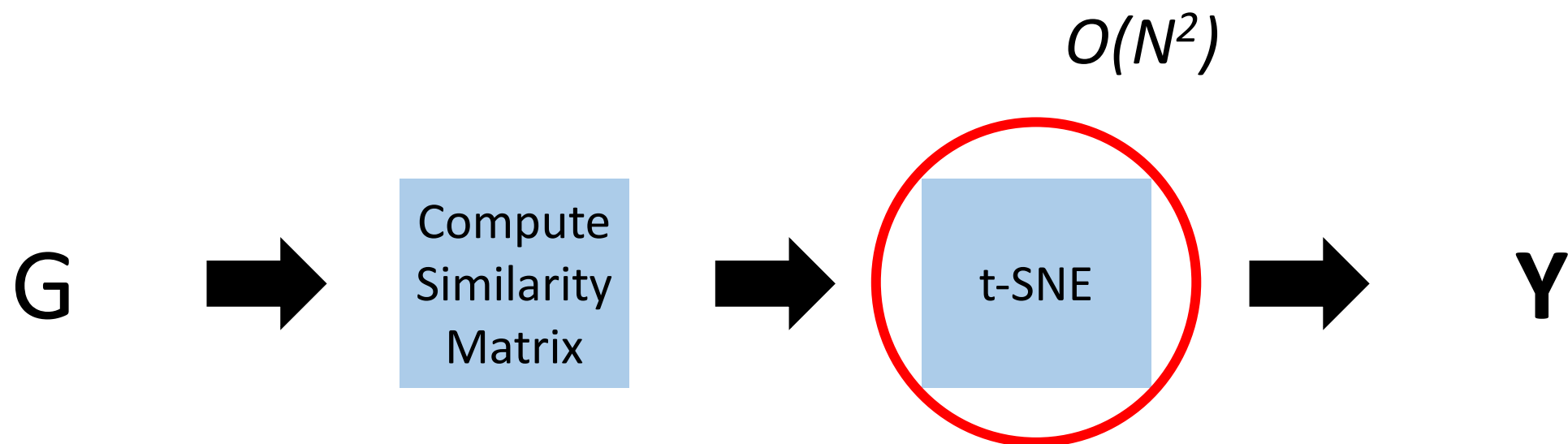
Faster Methods

- PMDS [1]
- SFDP [2], FM³ [3]
- CoReGD [4]
- DRGraph [5]



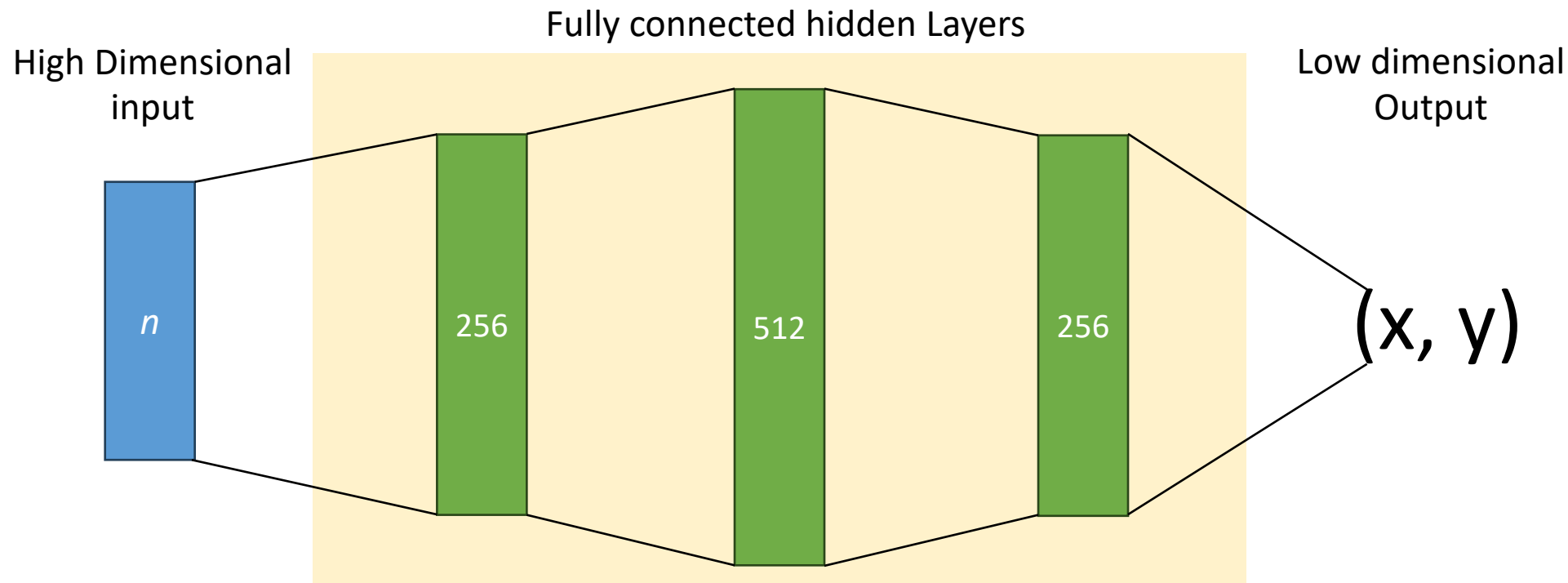
- 1 Ulrik Brandes and Christian Pich. **Eigensolver methods for progressive multidimensional scaling of large data**. In *Graph Drawing*, pages 42–53. Springer, 2007.
- 2 Yifan Hu. **Efficient, high-quality force-directed graph drawing**. *Mathematica journal*, 10(1):37– 71, 2005.
- 3 Martin Gronemann. **Engineering the fast-multipole-multilevel method for multicore and SIMD architectures**. Master’s thesis. Technische Univ. Dortmund, 2009.
- 4 Florian Grötschla, Joël Mathys, Robert Veres, and Roger Wattenhofer. **CoRe-GD: A Hierarchical Framework for Scalable Graph Visualization with GNNs**. In *ICLR*, Vienna, Austria, May 2024.
- 5 Minfeng Zhu, Wei Chen, Yuanzhe Hu, Yuxuan Hou, Liangjun Liu, and Kaiyuan Zhang. **DRGraph: An efficient graph layout algorithm for large-scale graphs by dimensionality reduction**. *IEEE TVCG*, 27(2):1666–1676, 2020.

tsNET



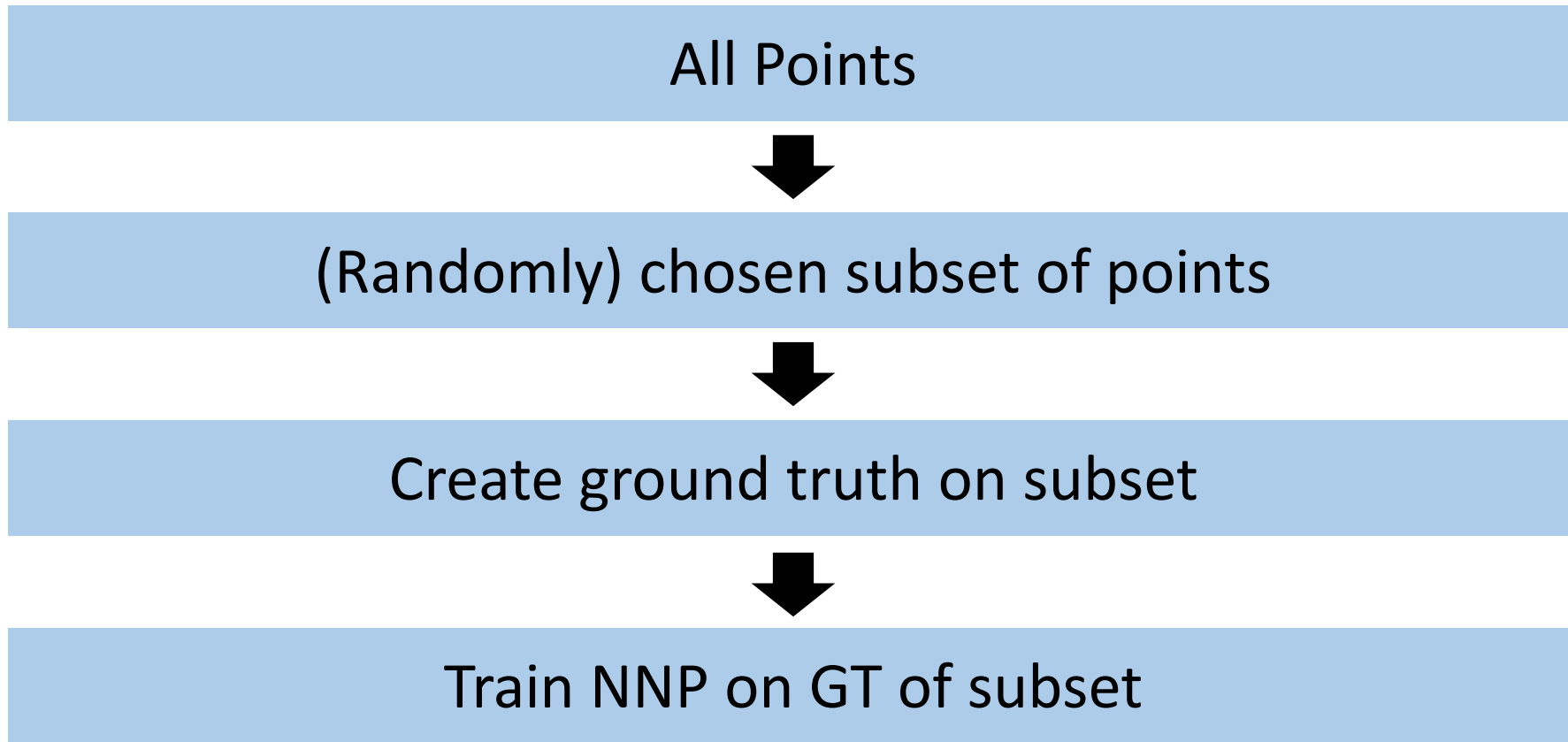
NNP (Neural Network Projections) [6]

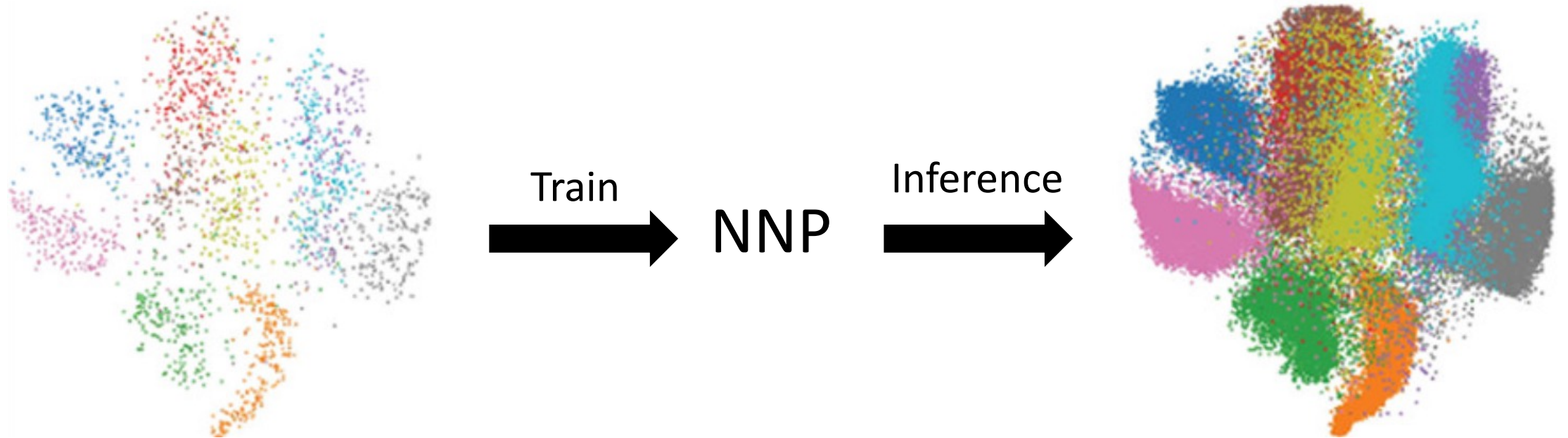
- Linear time complexity
- Can imitate any Dimensionality Reduction method



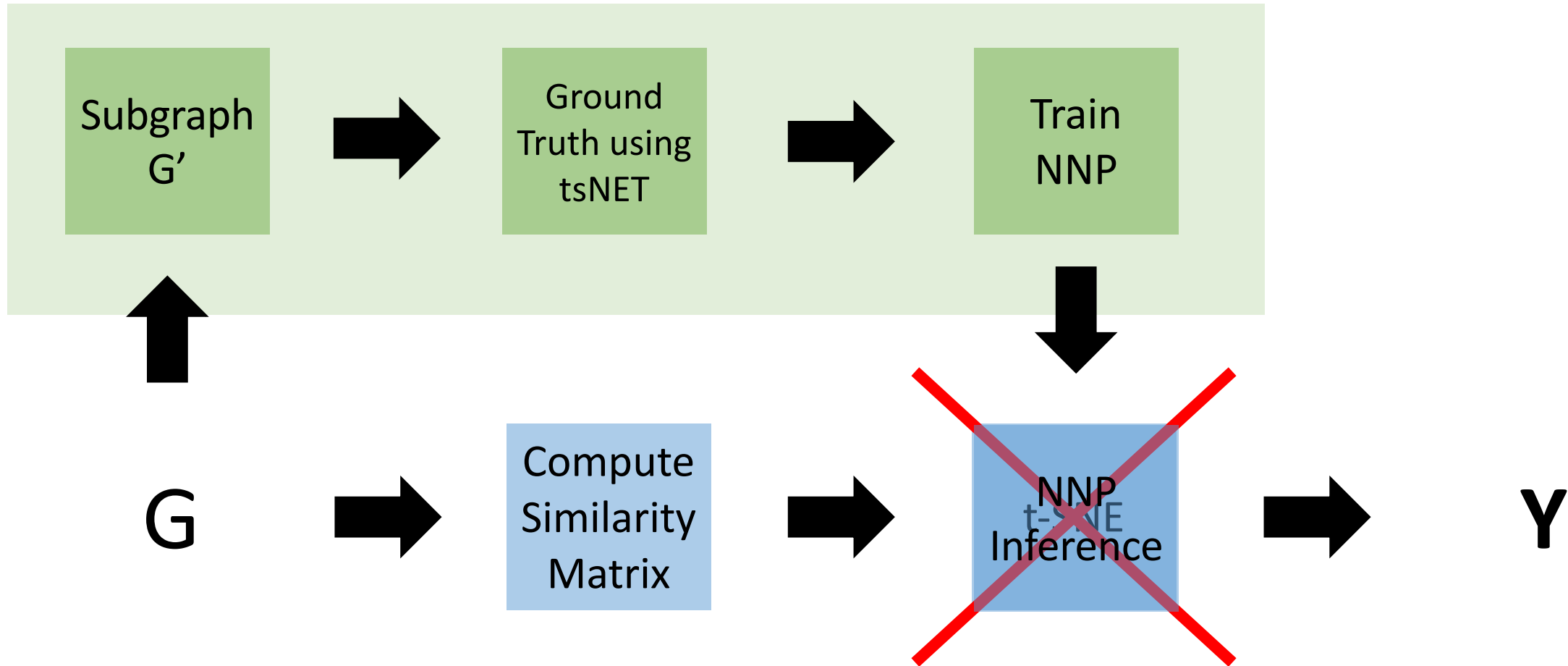
6. Mateus Espadoto, Nina Hirata, Tomita Sumiko, and Alexandru C. Telea. **Deep learning multidimensional projections.** *Information Visualization*, 19(3):247–269, 2020

NNP Training

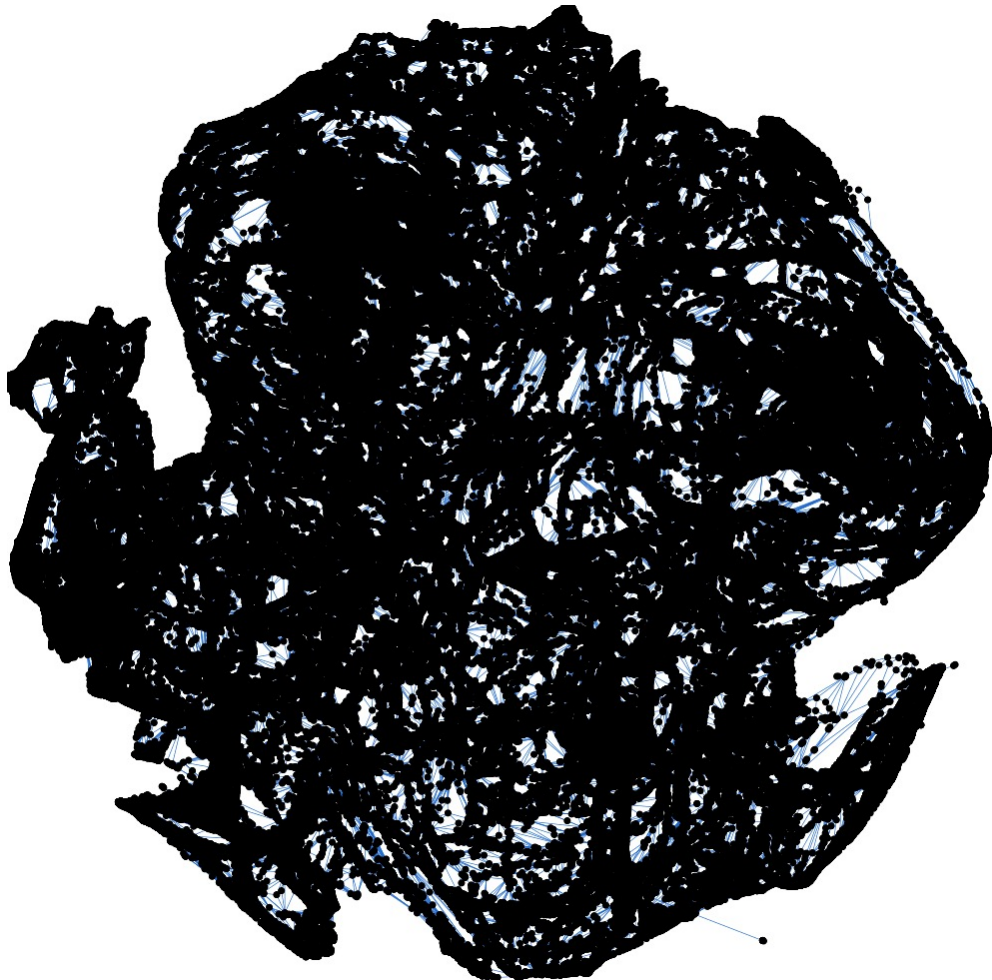




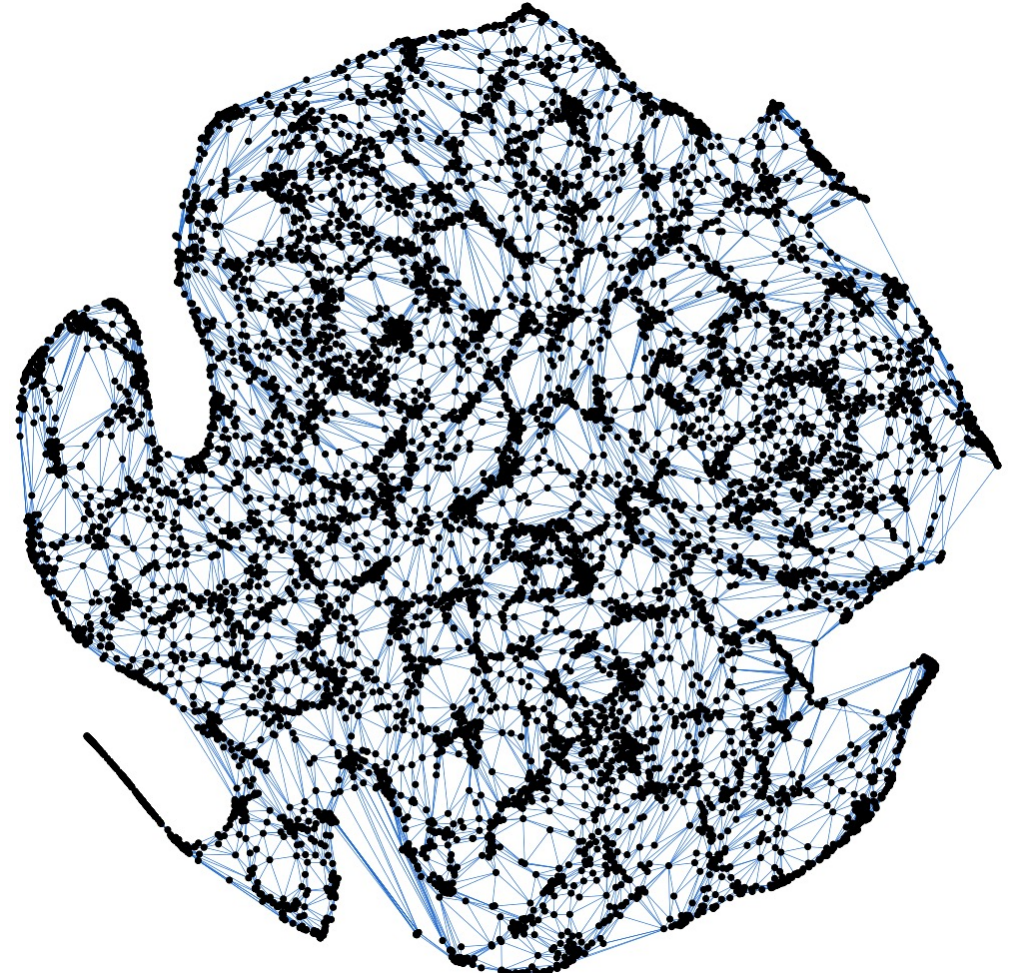
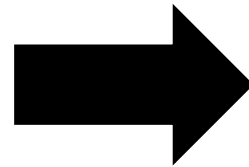
NNP-NET



Graph Coarsening

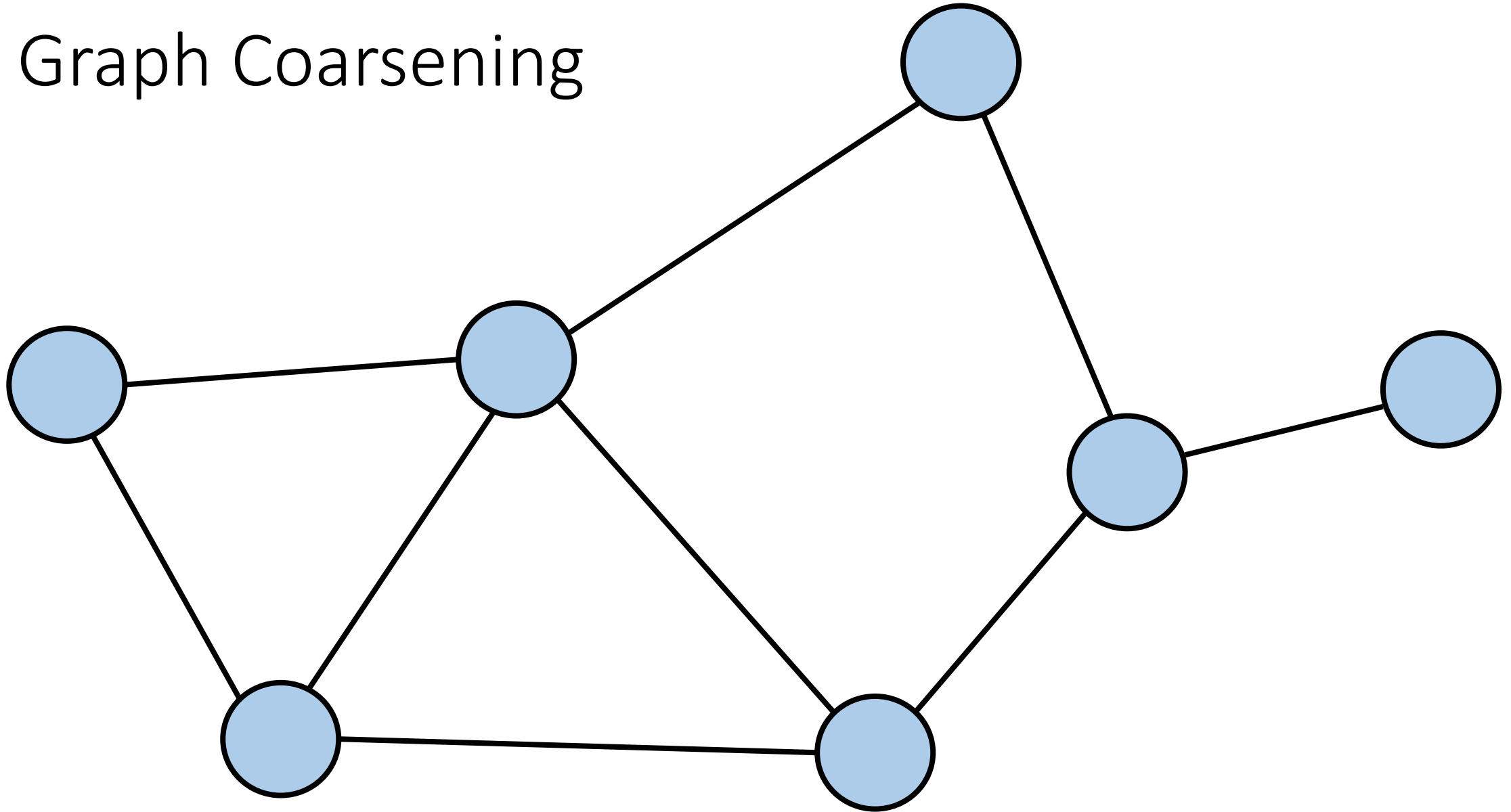


Nodes: 914 231

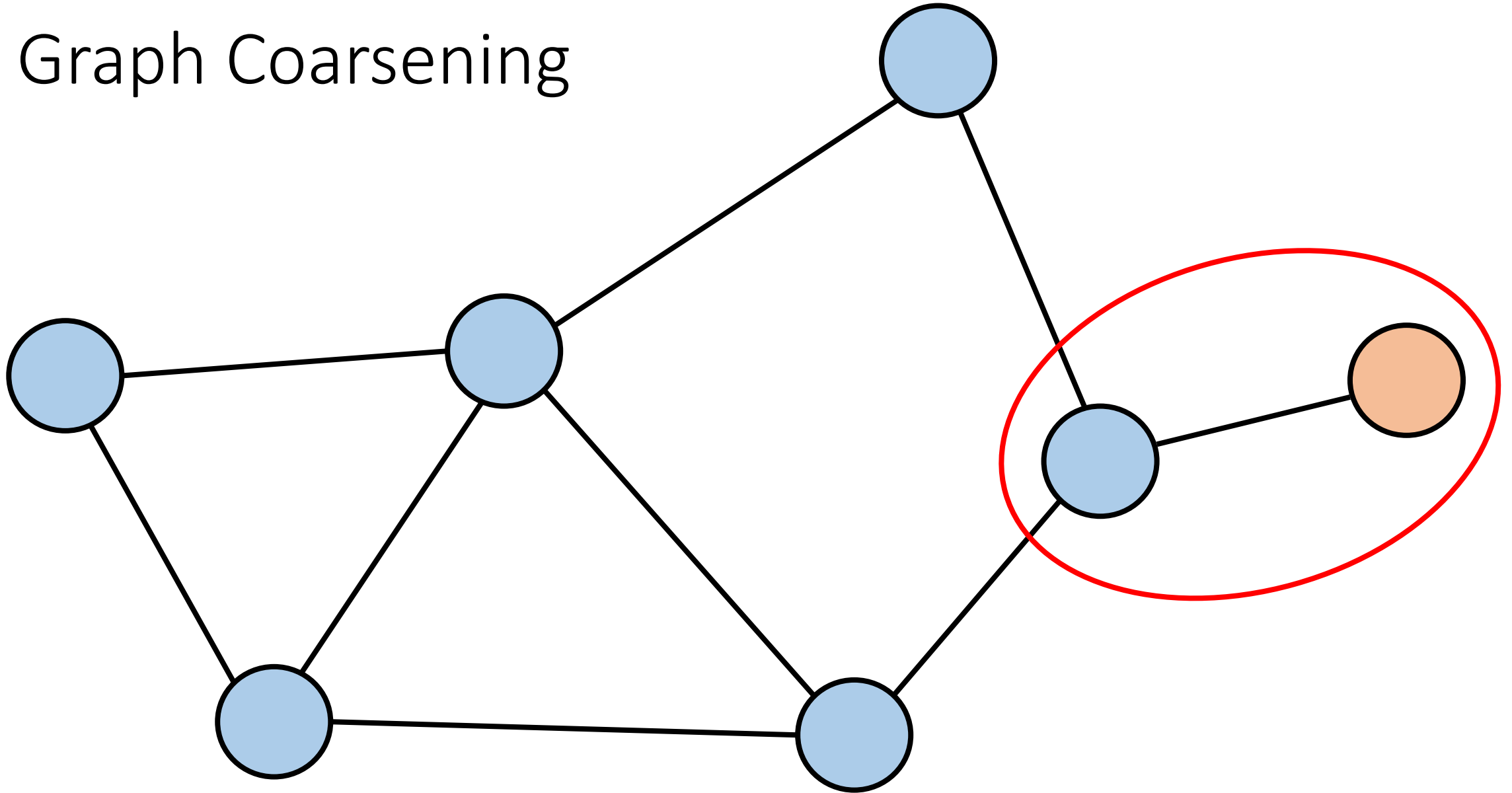


Nodes: 10 000

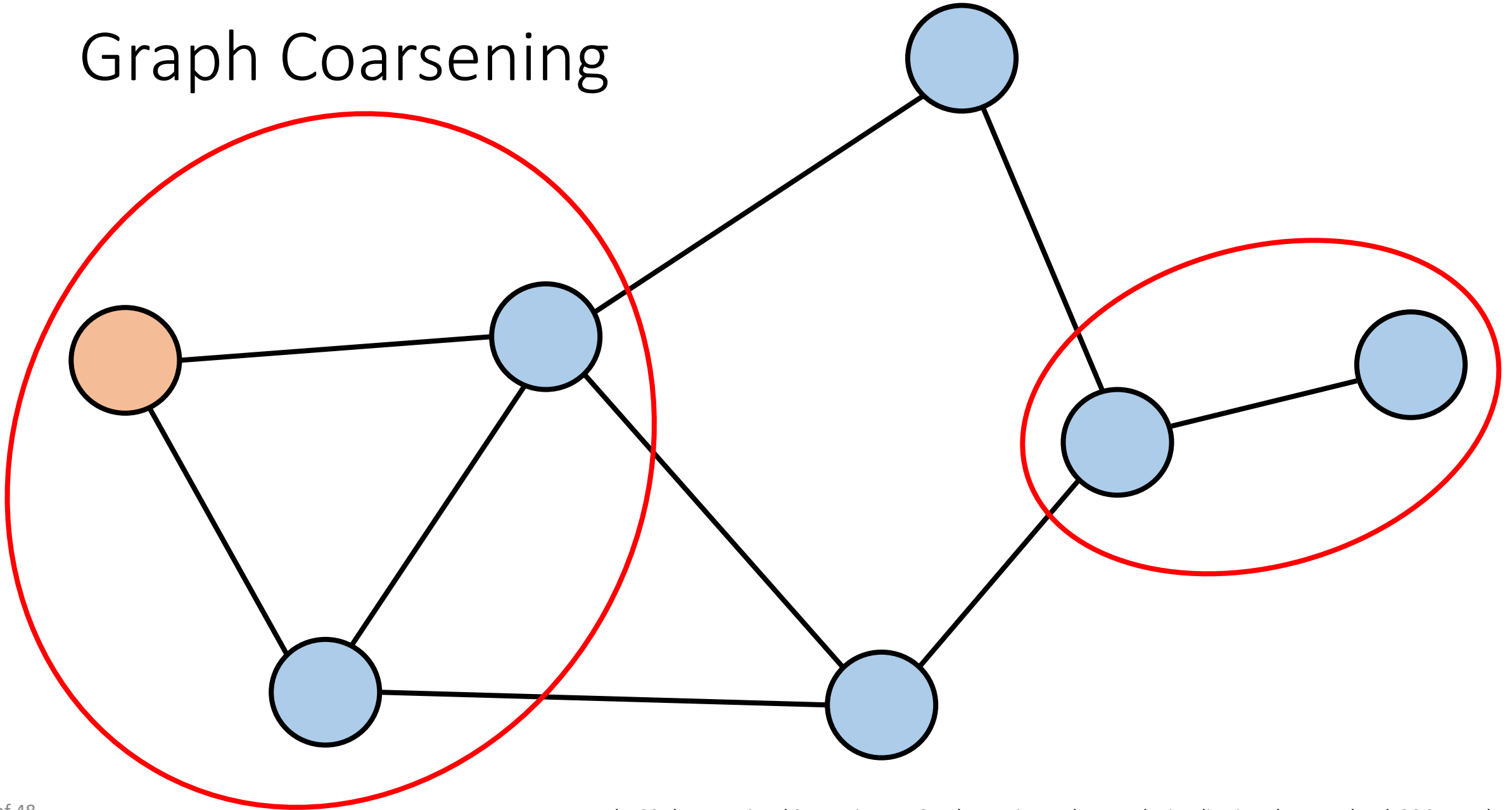
Graph Coarsening



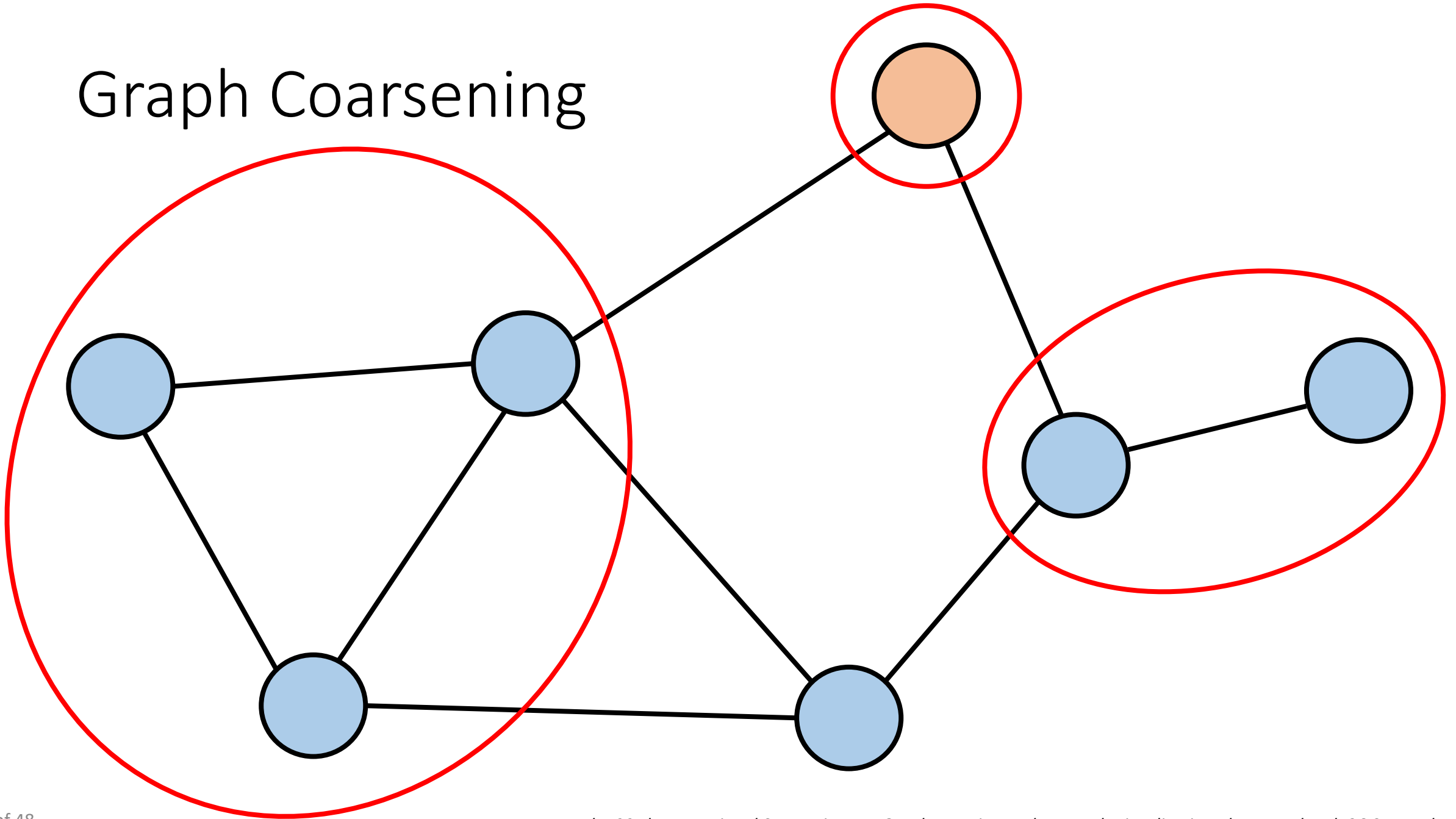
Graph Coarsening



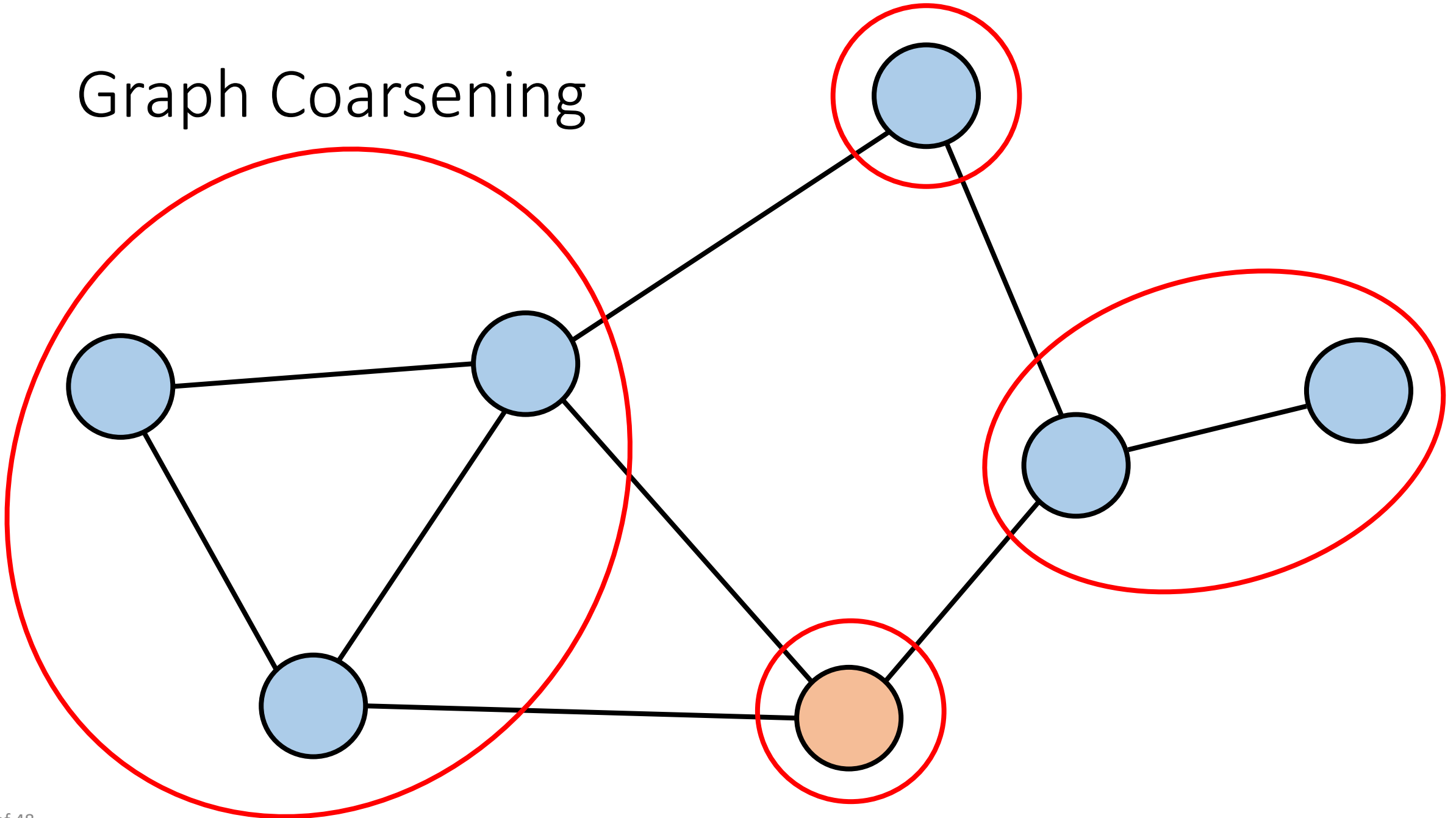
Graph Coarsening



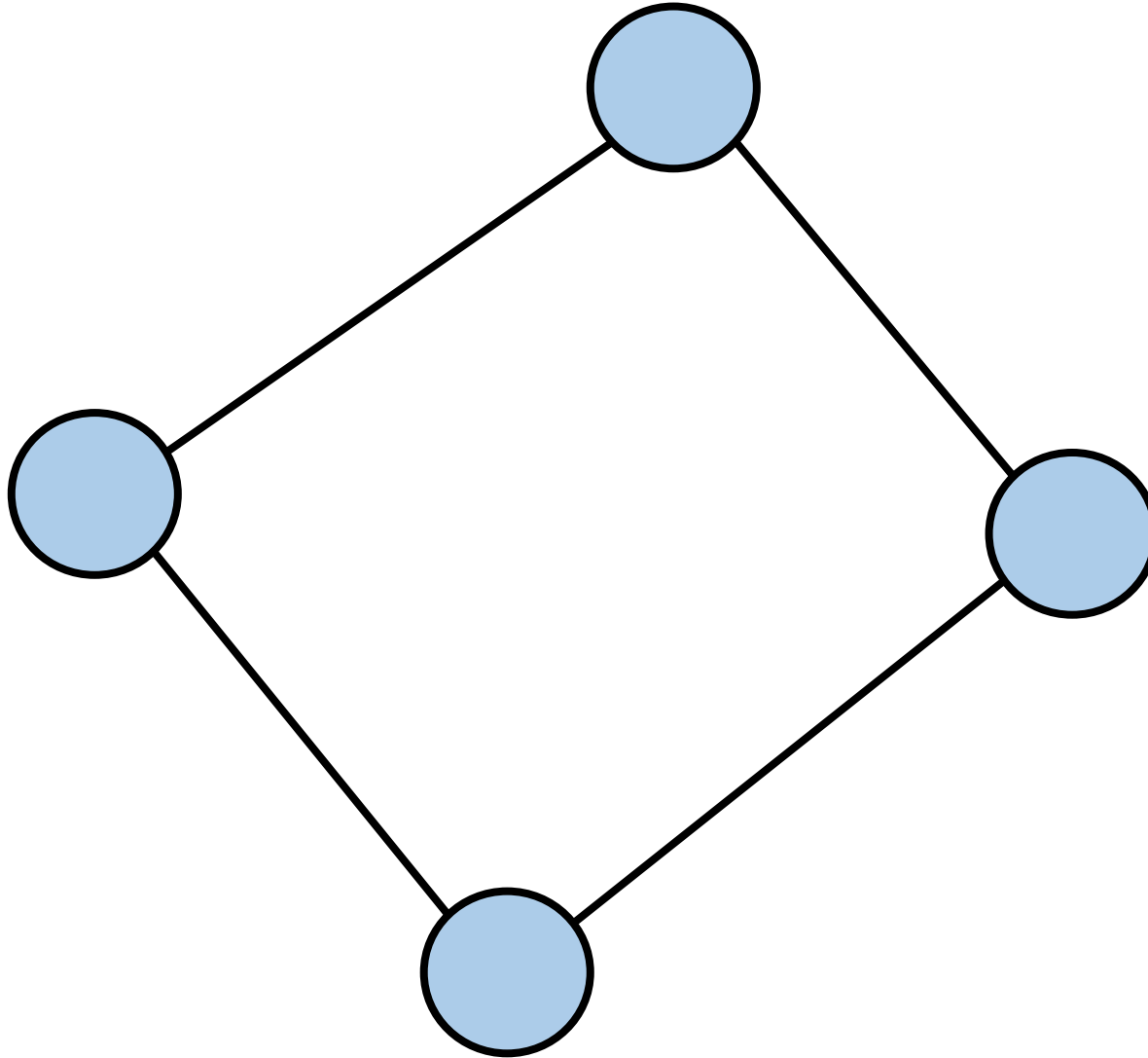
Graph Coarsening



Graph Coarsening



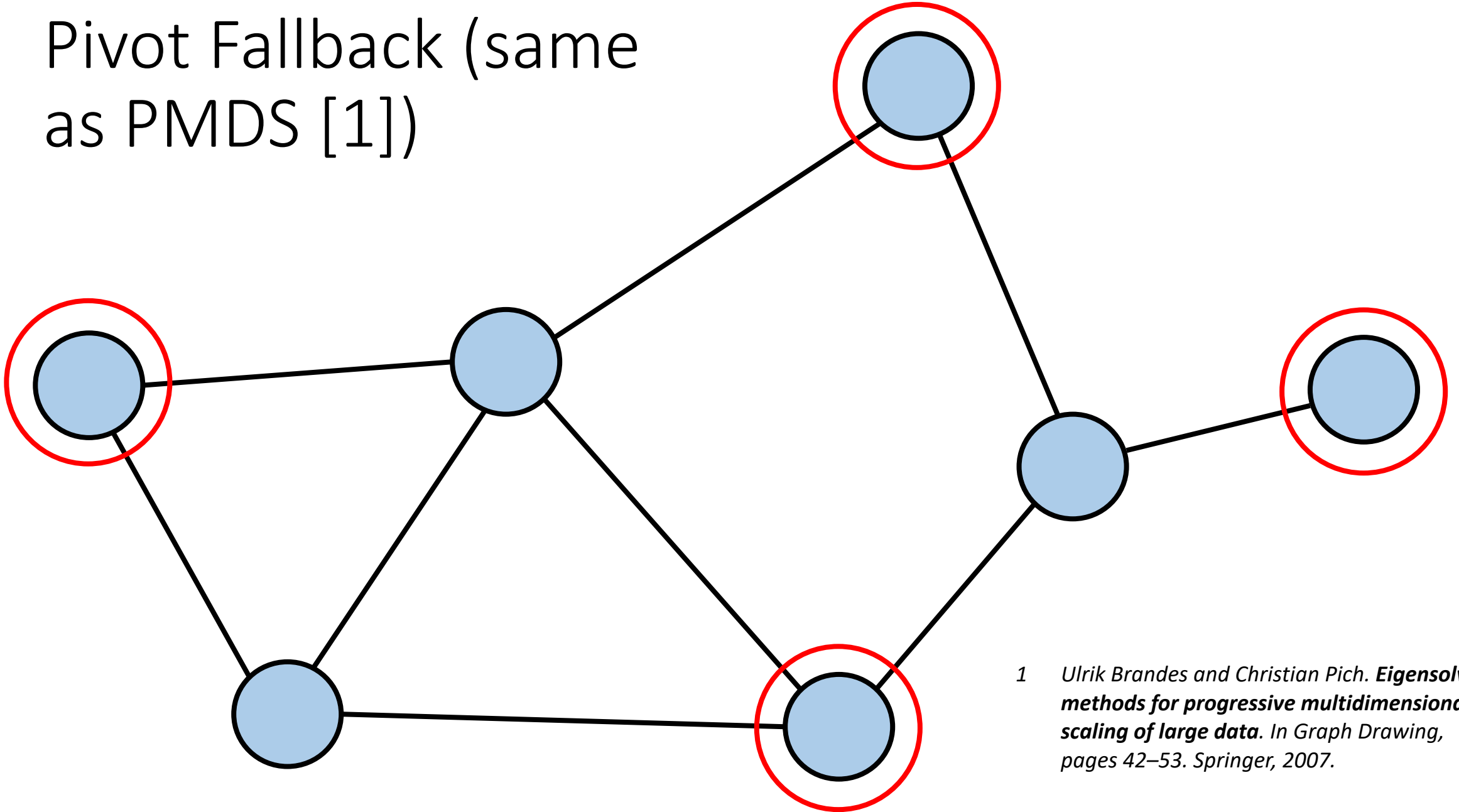
Graph Coarsening



Graph Coarsening

- Very fast (< 1 second for graphs with 1 million nodes)
- Struggles with some graphs

Pivot Fallback (same as PMDS [1])

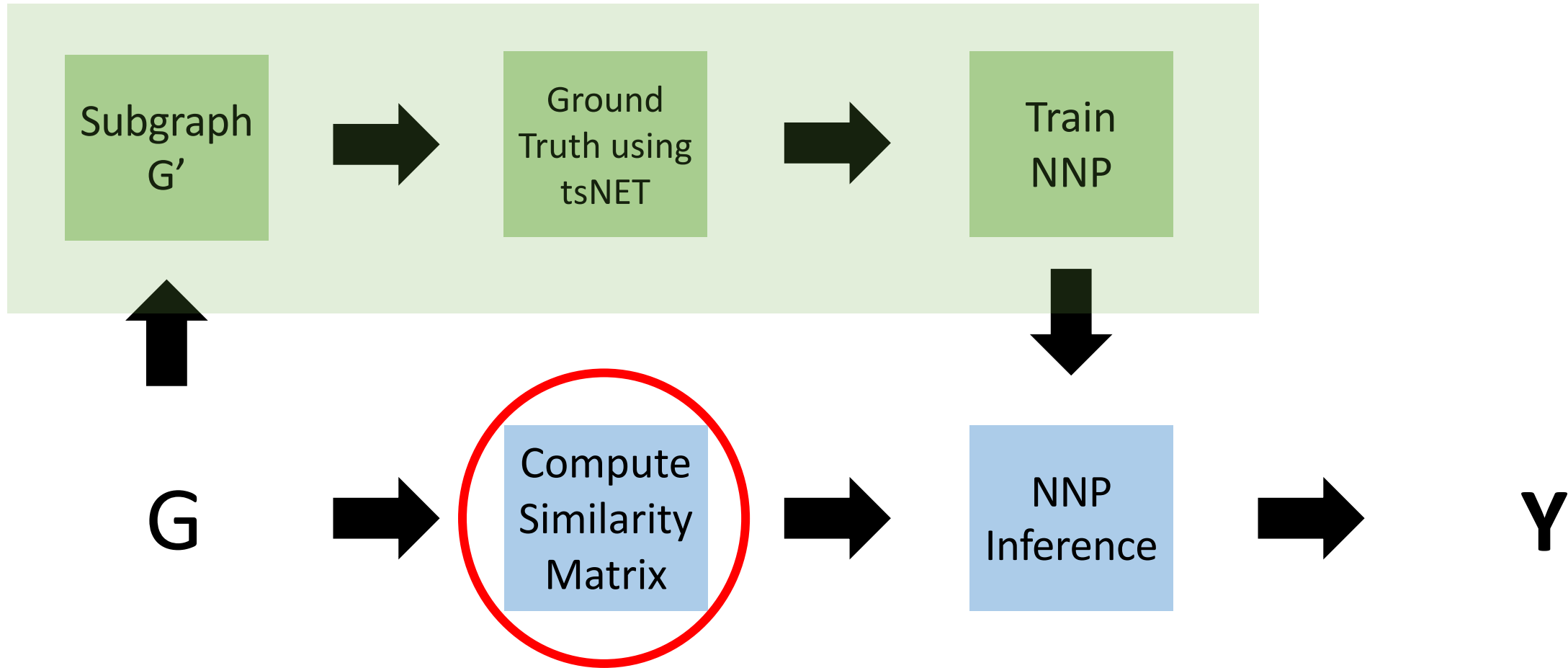


- 1 Ulrik Brandes and Christian Pich. **Eigensolver methods for progressive multidimensional scaling of large data**. In *Graph Drawing*, pages 42–53. Springer, 2007.

Pivot Fallback

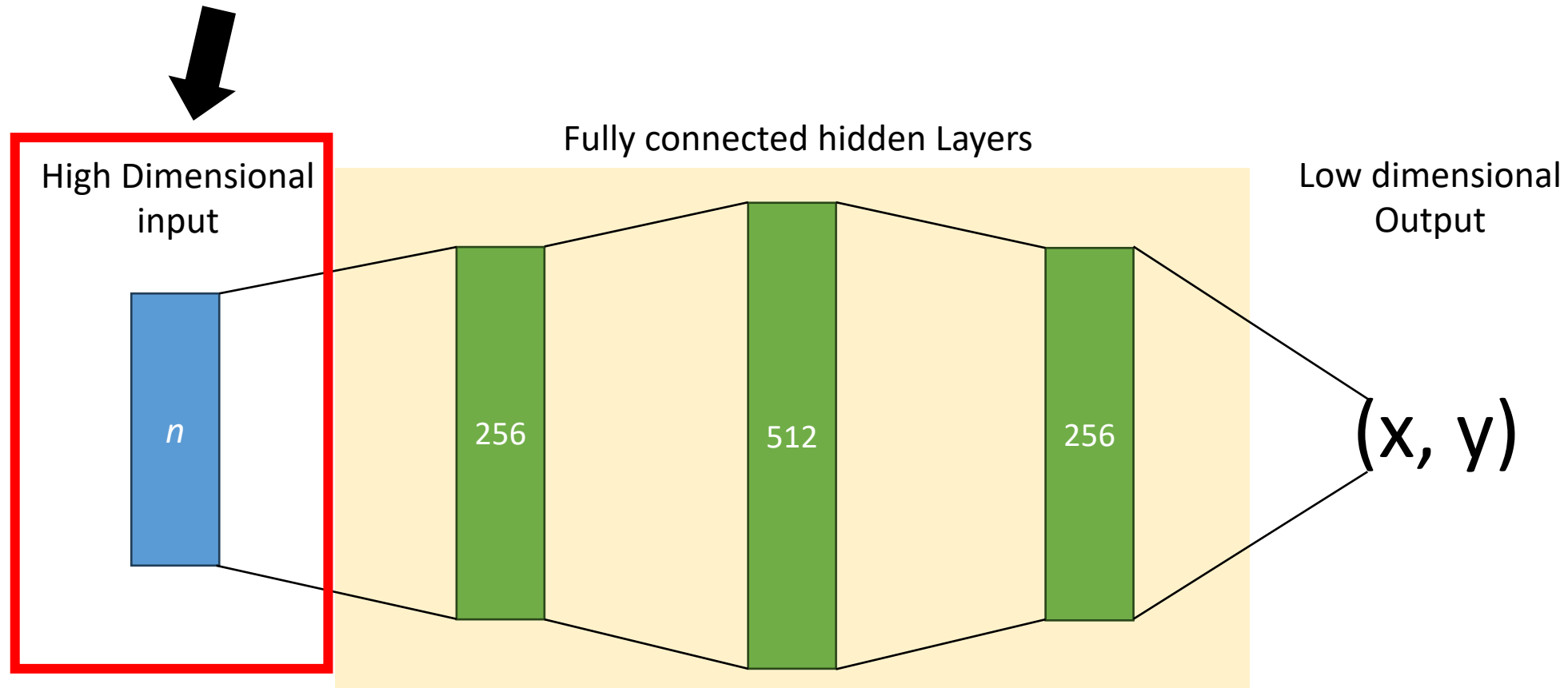
- Always gives good results
- Slow

NNP-NET

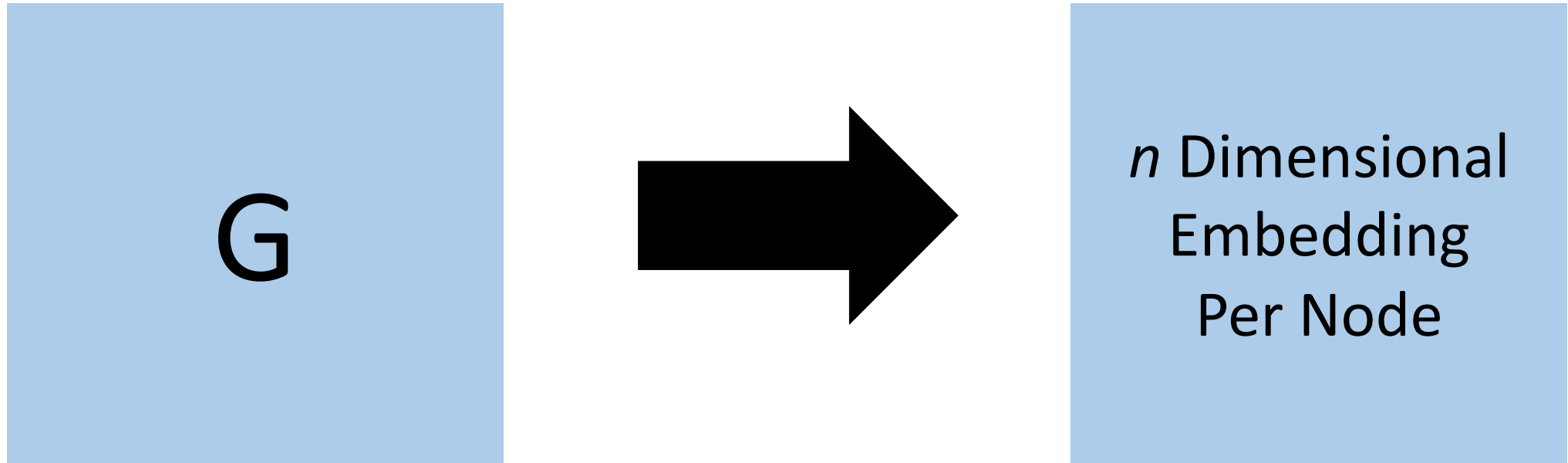


Input Problem

We don't have this.



What we need:



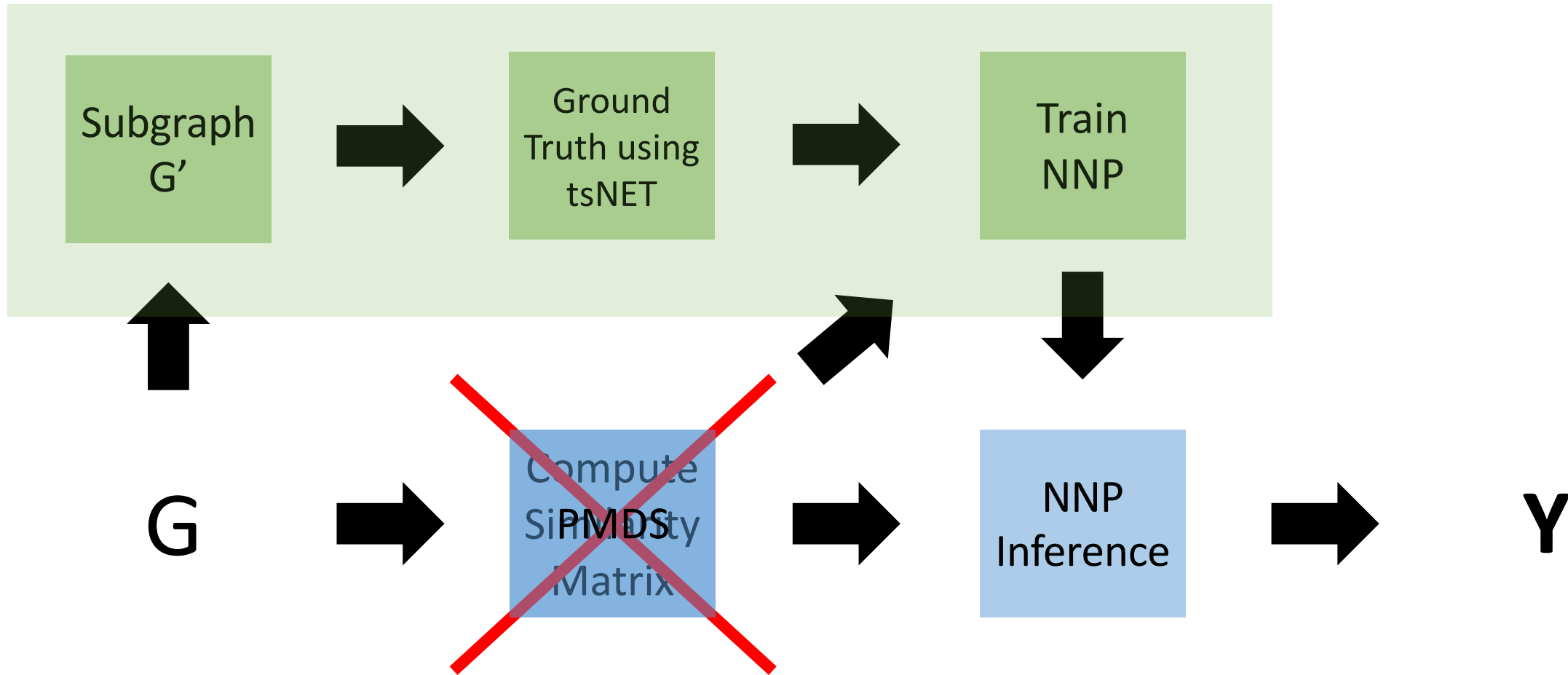
*What can create an n
Dimensional Embedding
Per Node From a Graph G ?*

Graph Drawing!
(PMDS in our case)

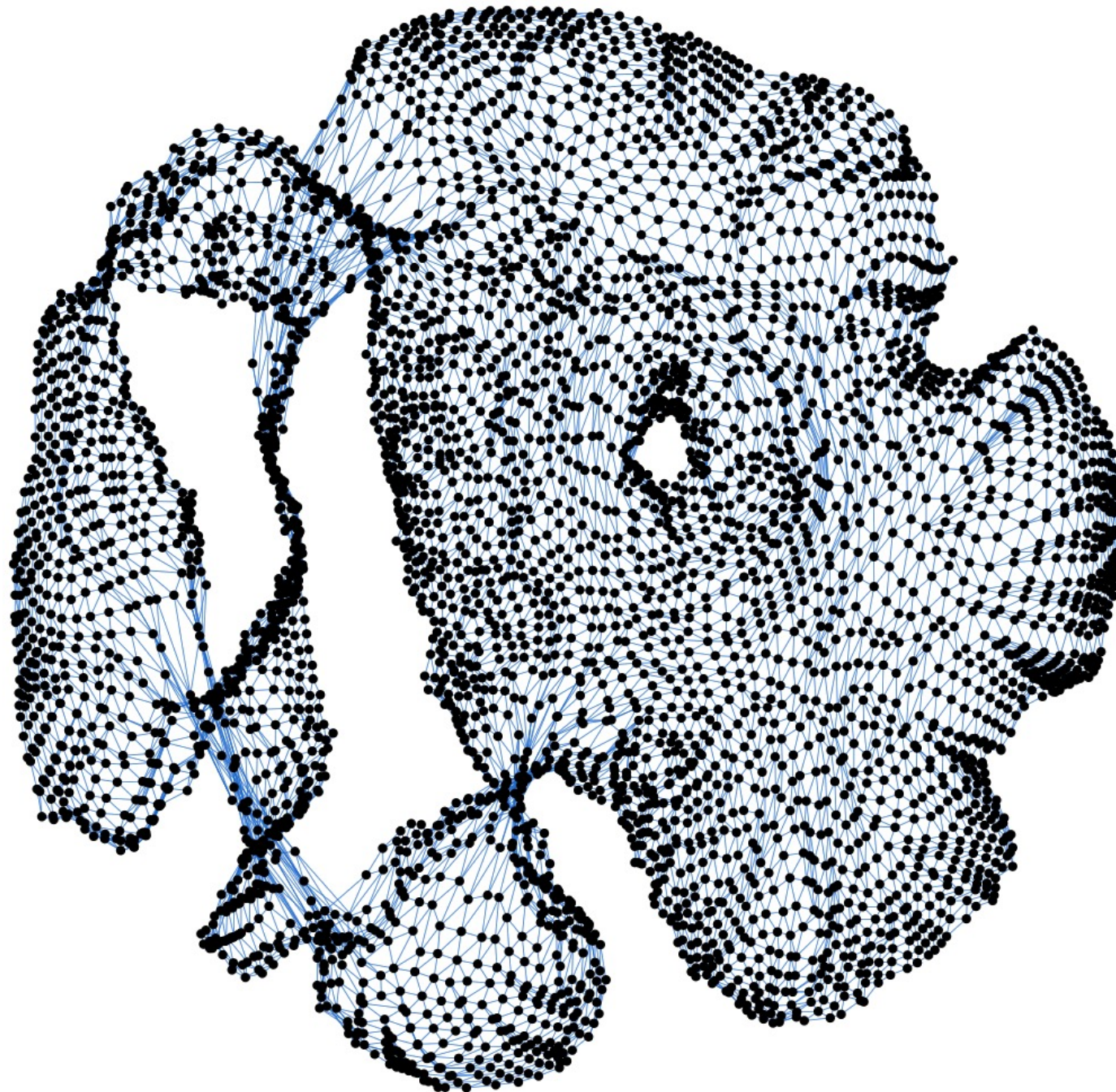
The Idea:

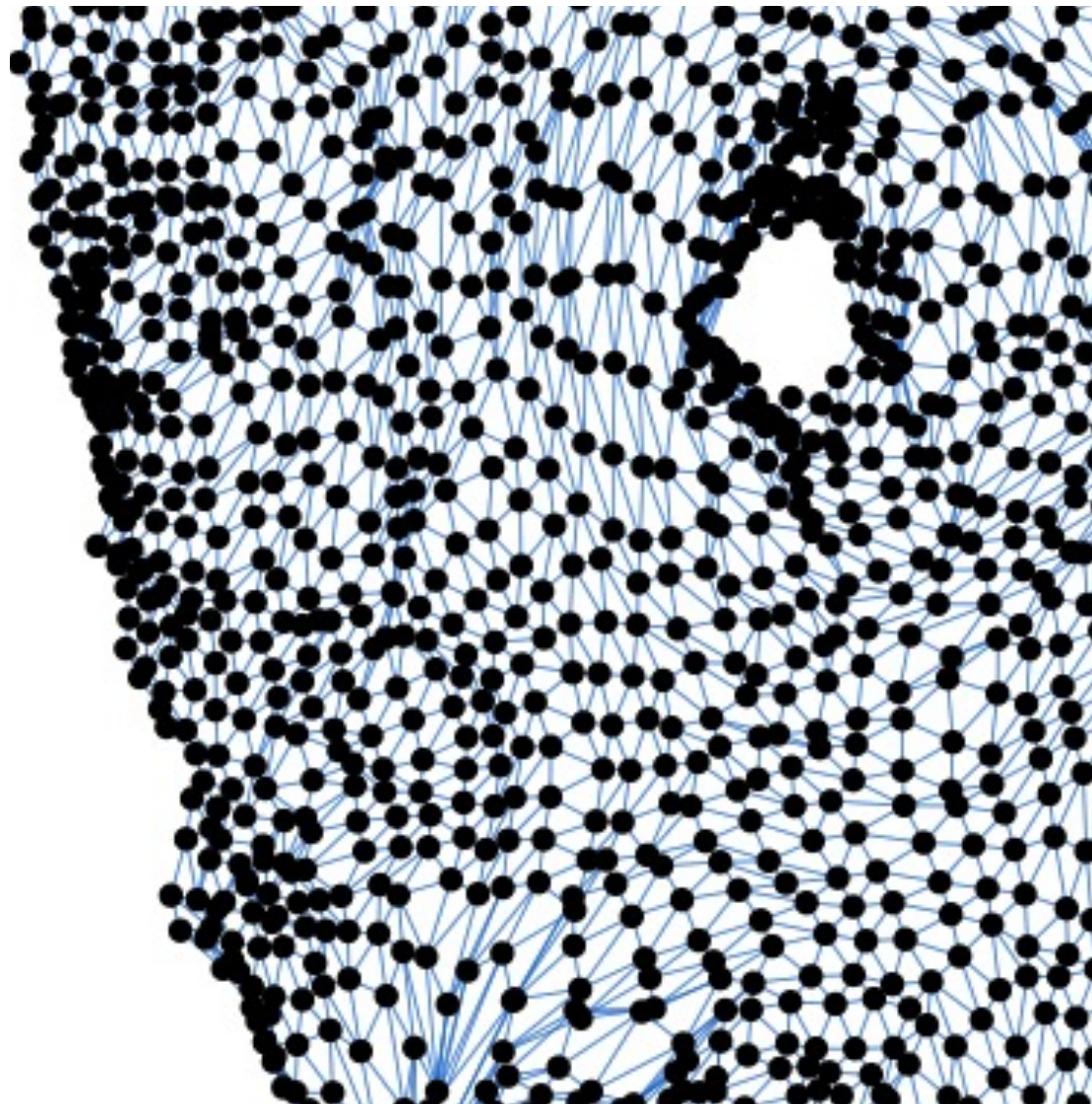
- Create 50-dimensional layout using PMDS
- Reduce 50-dimensional layout to 2 using NNP

NNP-NET

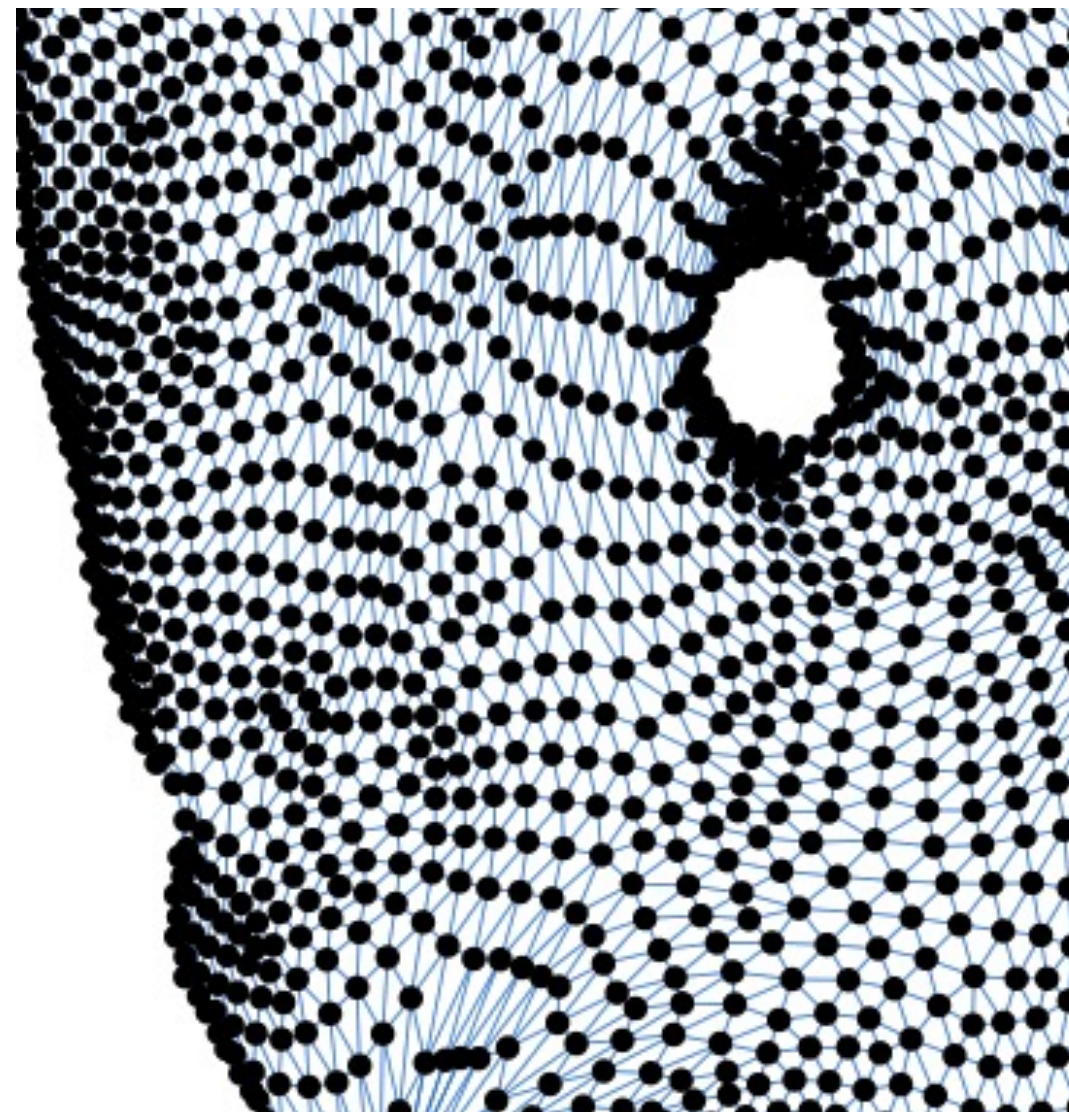


Results

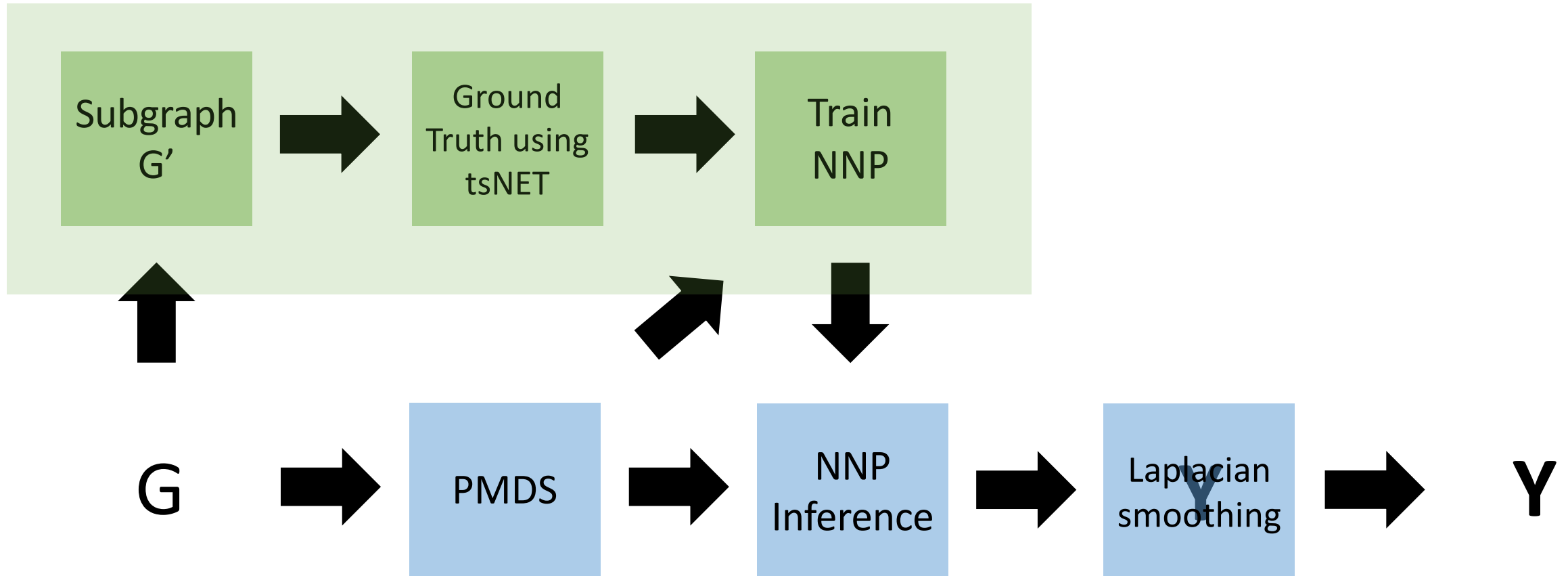




Laplacian
Smoothing

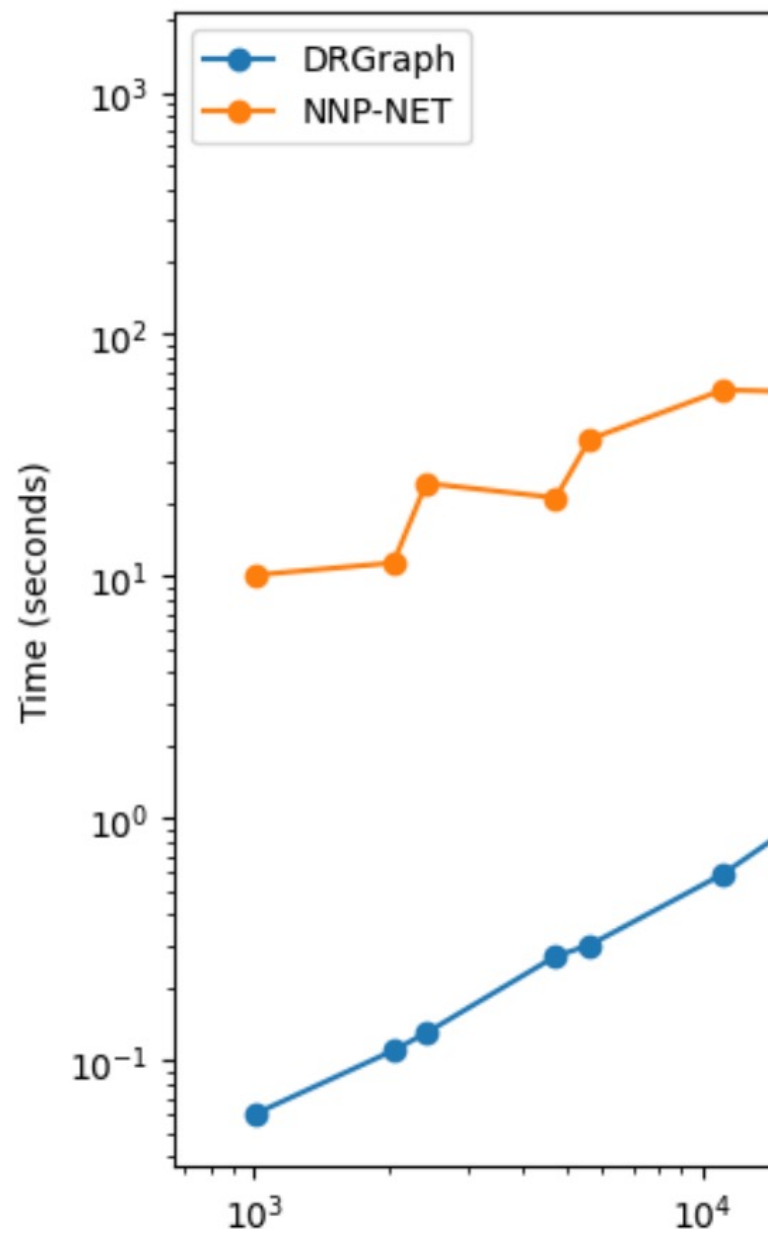


NNP-NET



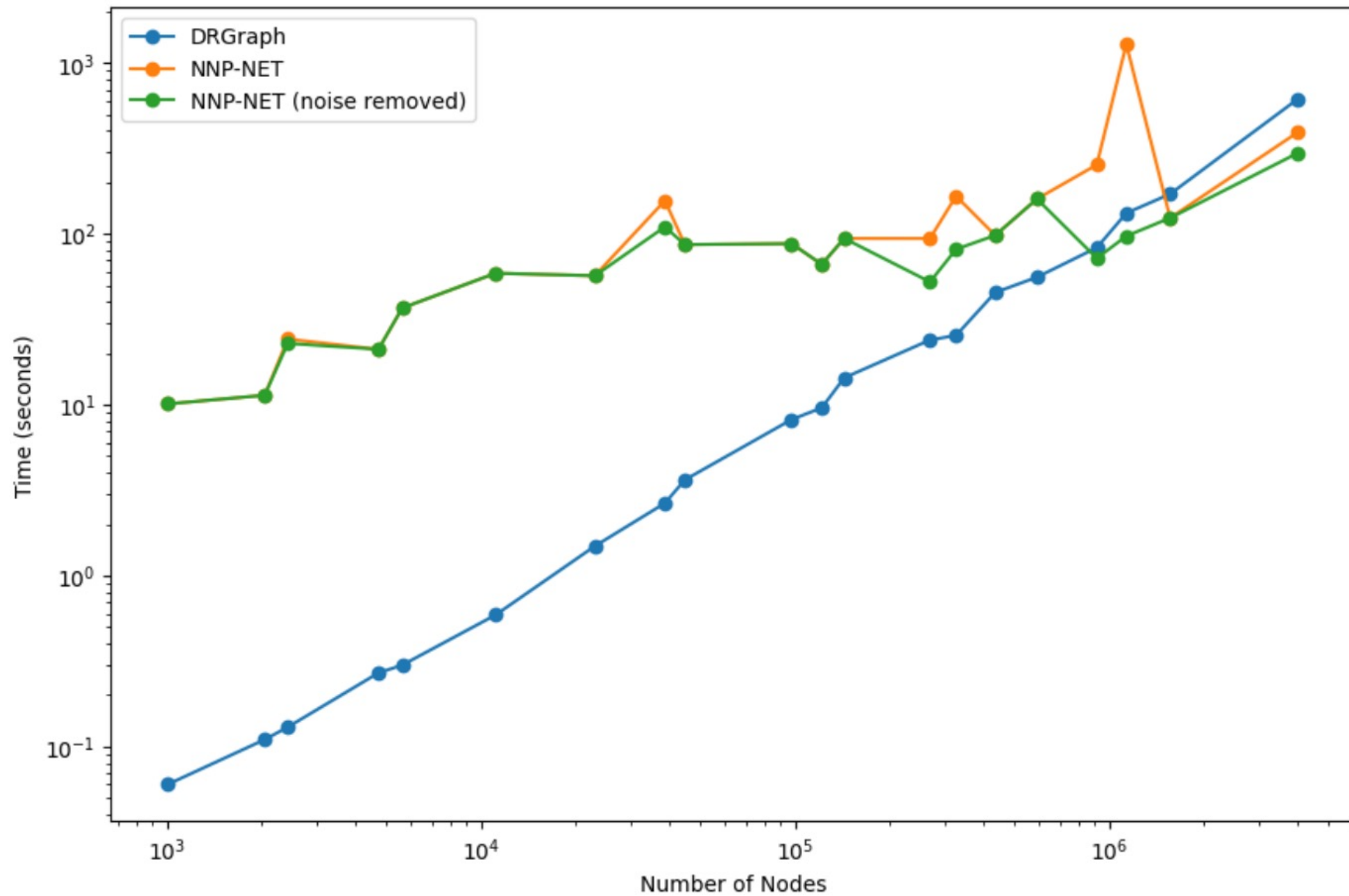
Results

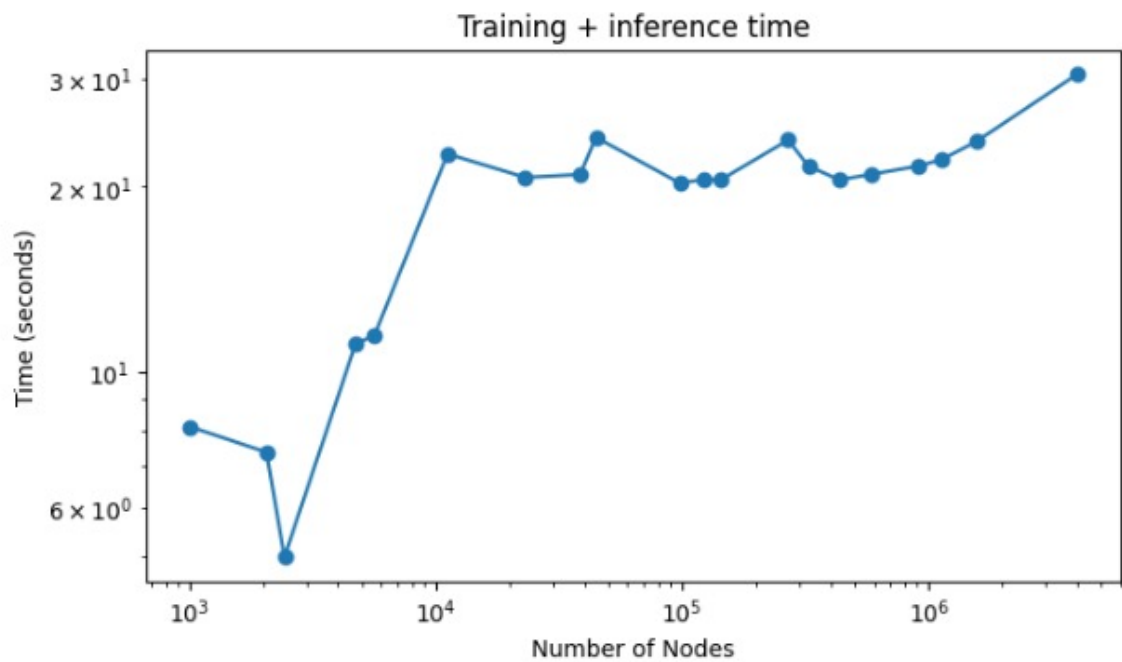
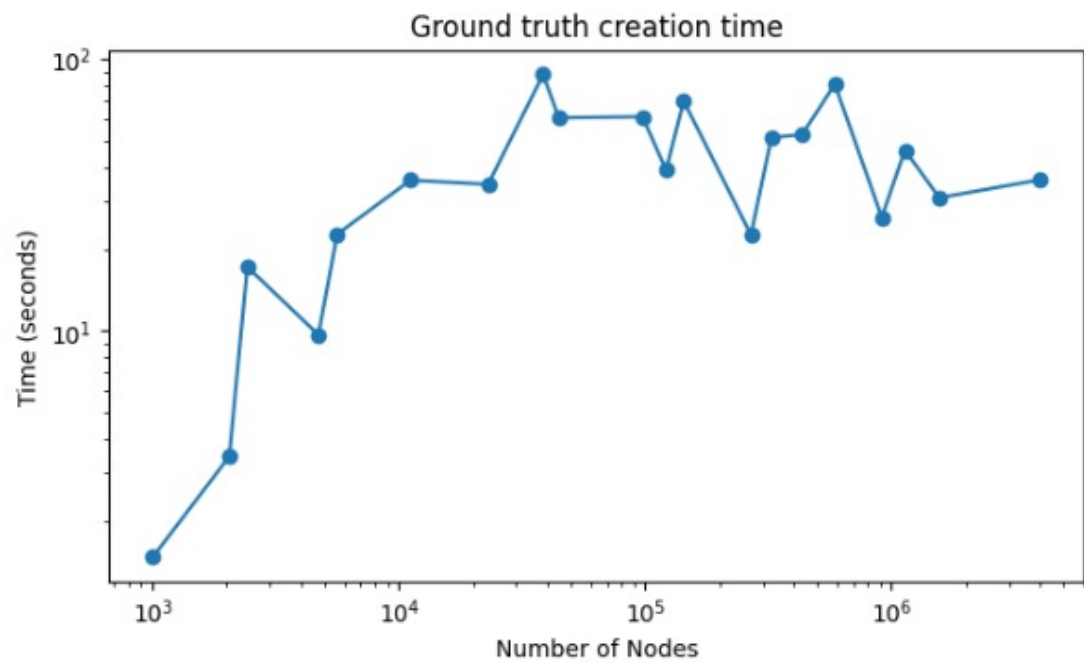
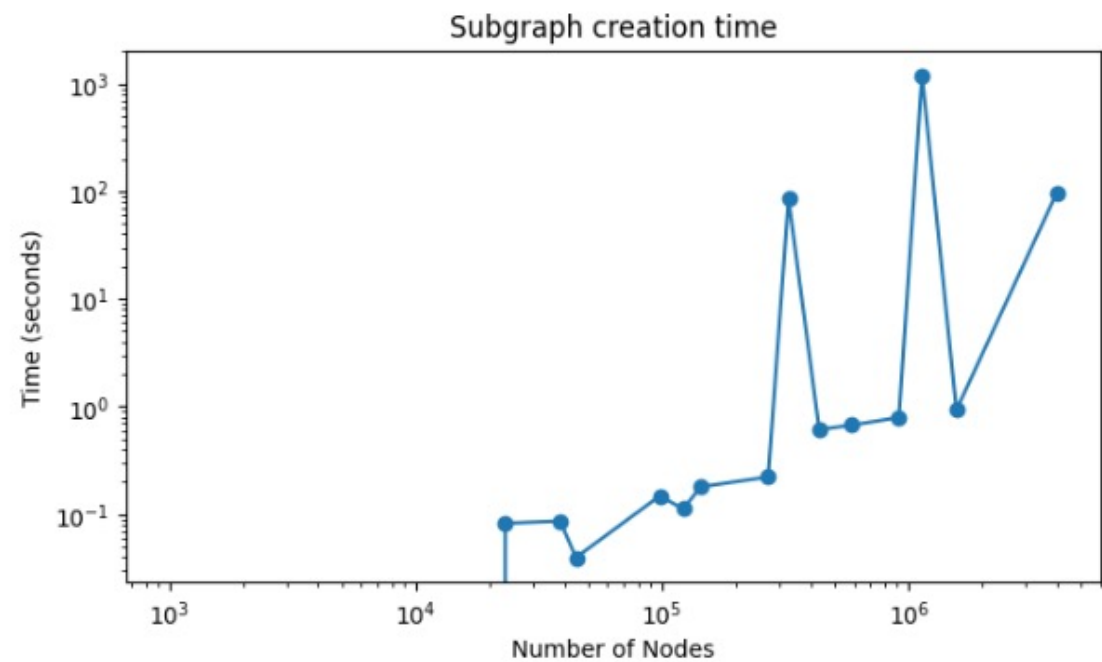
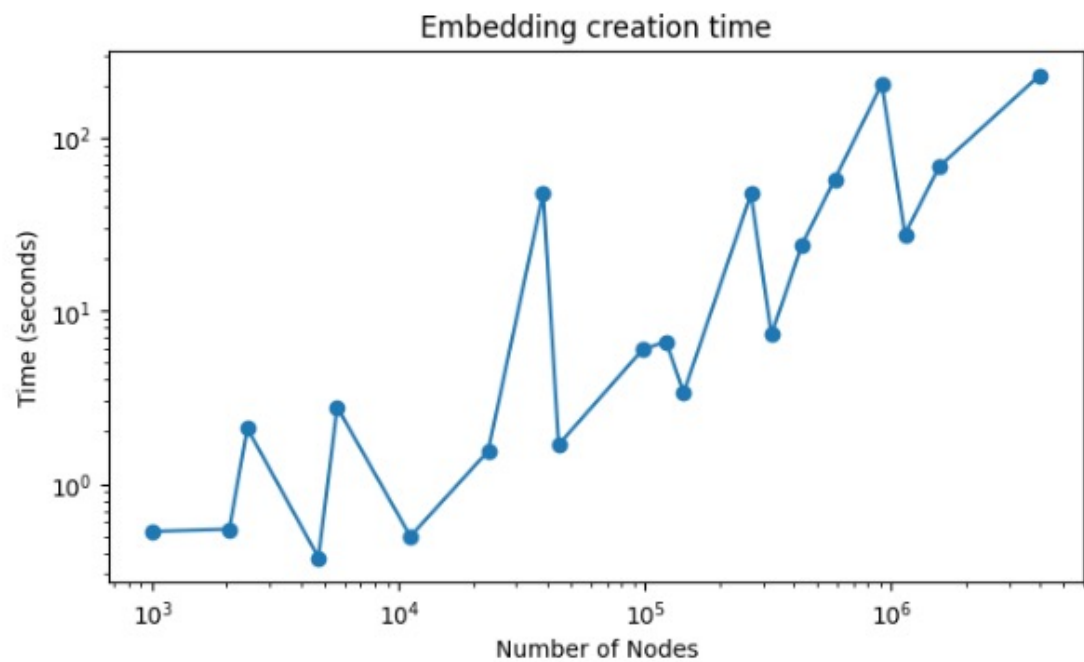
- Execution time compared to DRGraph
 - Timings of each individual part
- Visual comparison
- Quantitative comparison using metrics
 - Neighborhood preservation
 - Stress

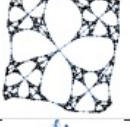
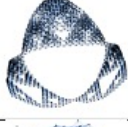
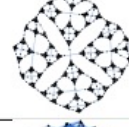
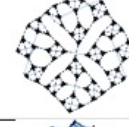
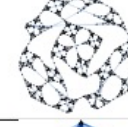
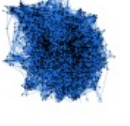
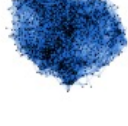
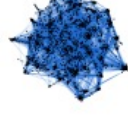
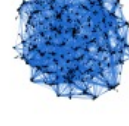
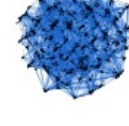
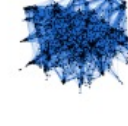
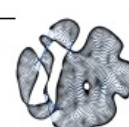
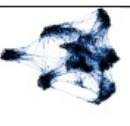
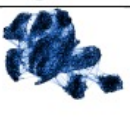
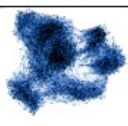
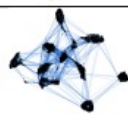
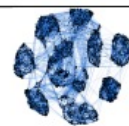
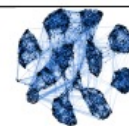
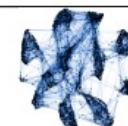
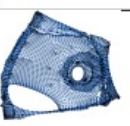
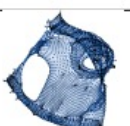
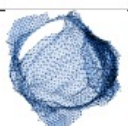
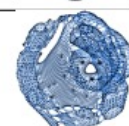
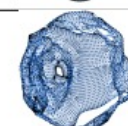
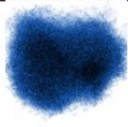
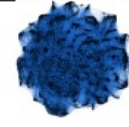
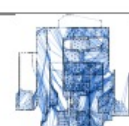
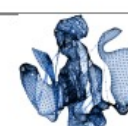
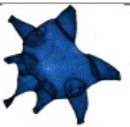
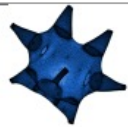
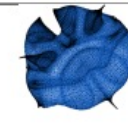


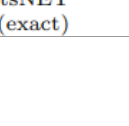
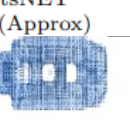
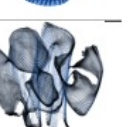
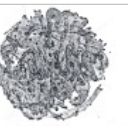
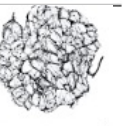
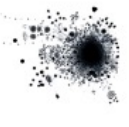
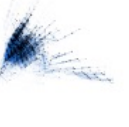
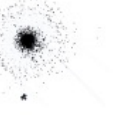
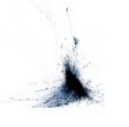
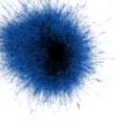
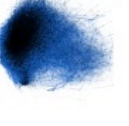
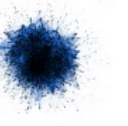
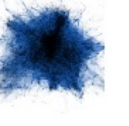
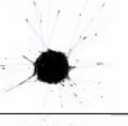
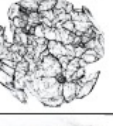
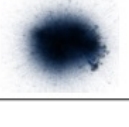
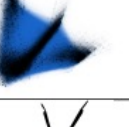
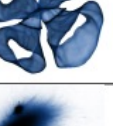
Performance Compared to DRGraph

- Rerun without Weights
- Remove pivot fallback times

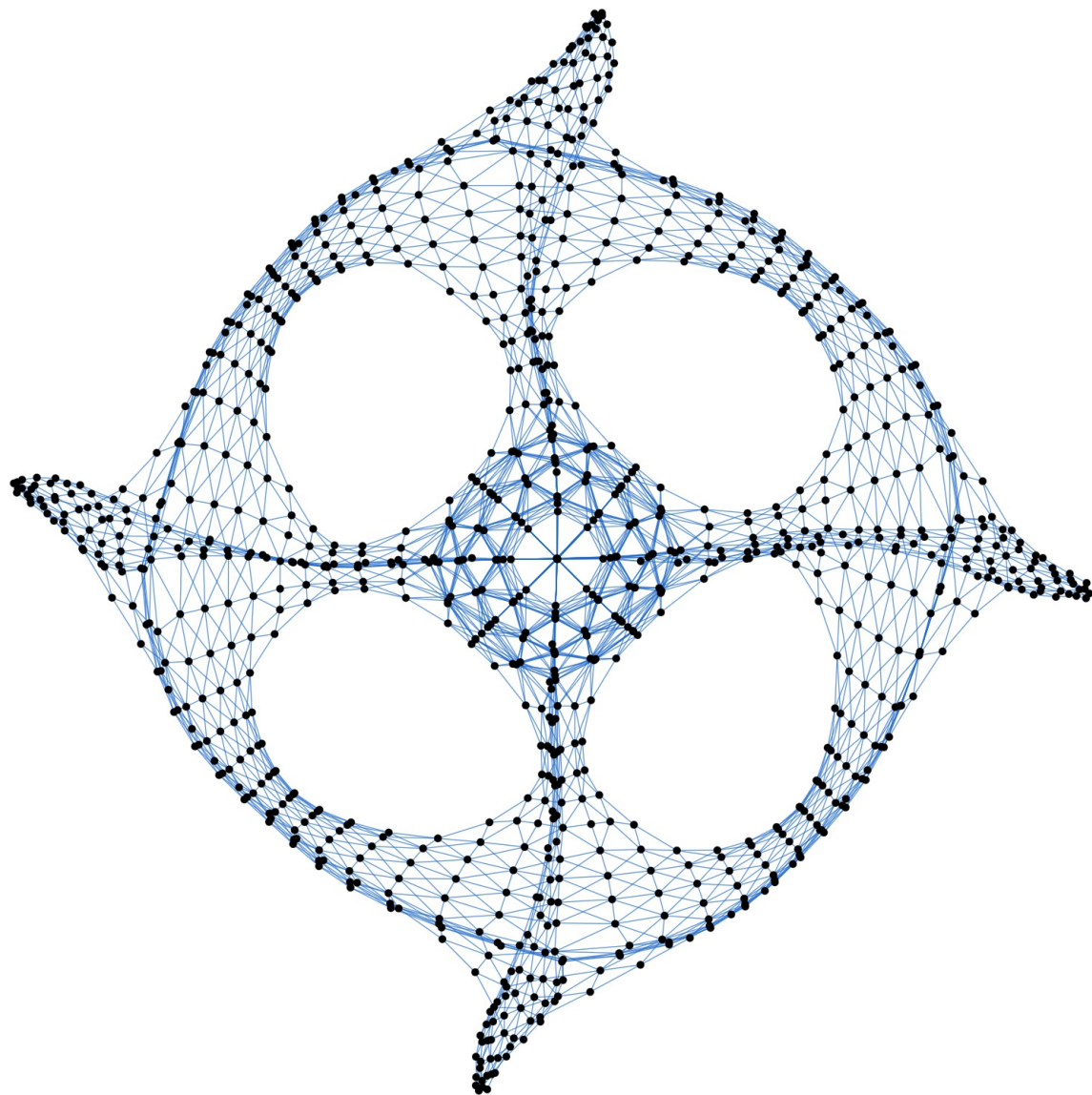




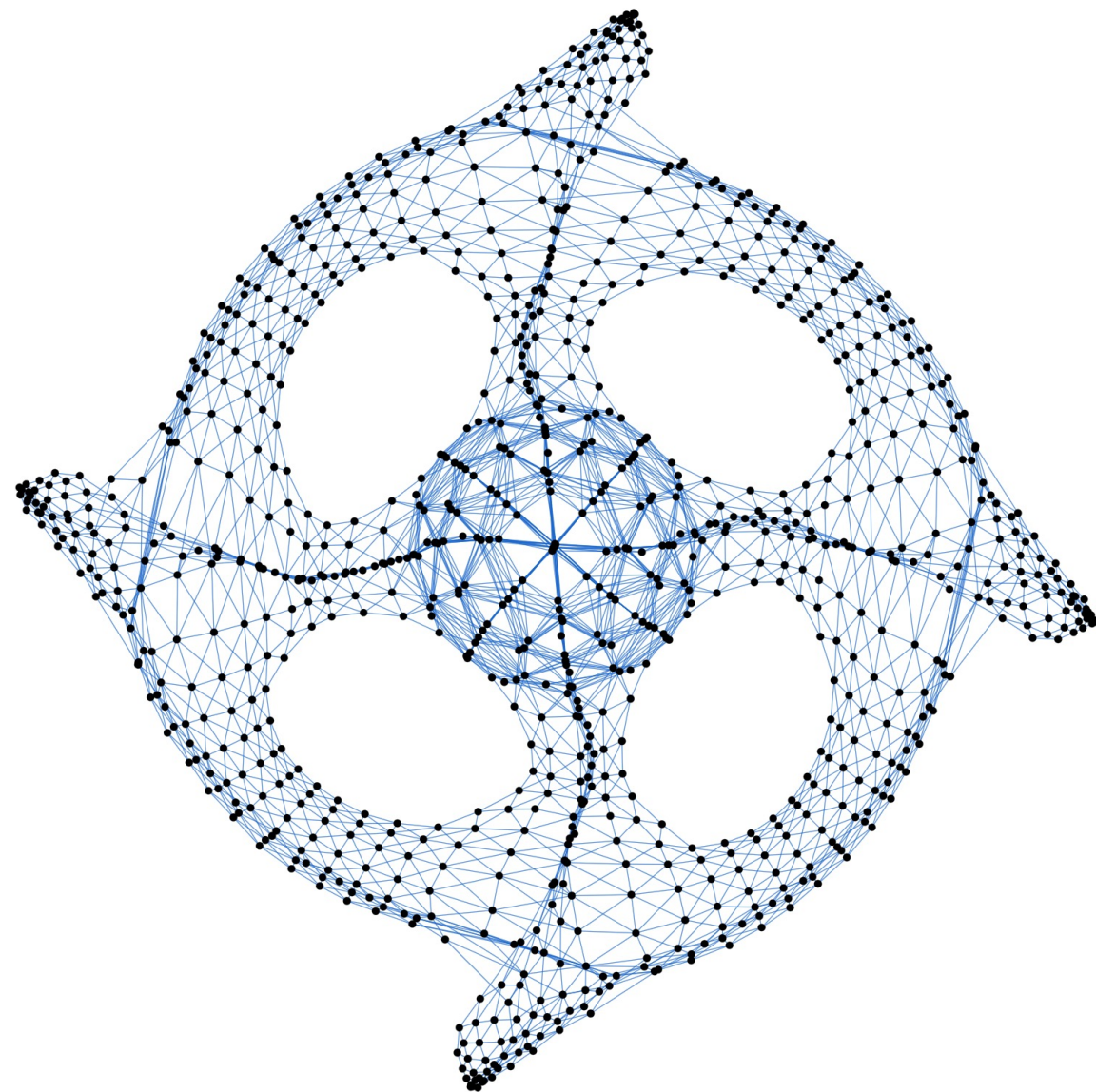
Graph	SFDP	FM ³	PMDS	DRGraph	tsNET* (exact)	tsNET* (Approx)	NNP-NET
dwt_1005							
sierpinski3d							
MIS-Knowledge-Map							
3elt							
optdigits_10NN							
fe_4elt2							
bcsstk36							
k49_norm_10NN							
fe_bcsstk32							
36 of 48 m_t1							

Graph	SFDP	FM ³	PMDS	DRGraph	tsNET* (exact)	tsNET* (Approx)	NNP-NET
							
ship_003							
fe_ocean							
ok2010							
web-NotreDame							
CoPapers-Citeseer							
gsm_106857							
tx2010							
com-Youtube							
Flan_1565							
com-LiveJournal							

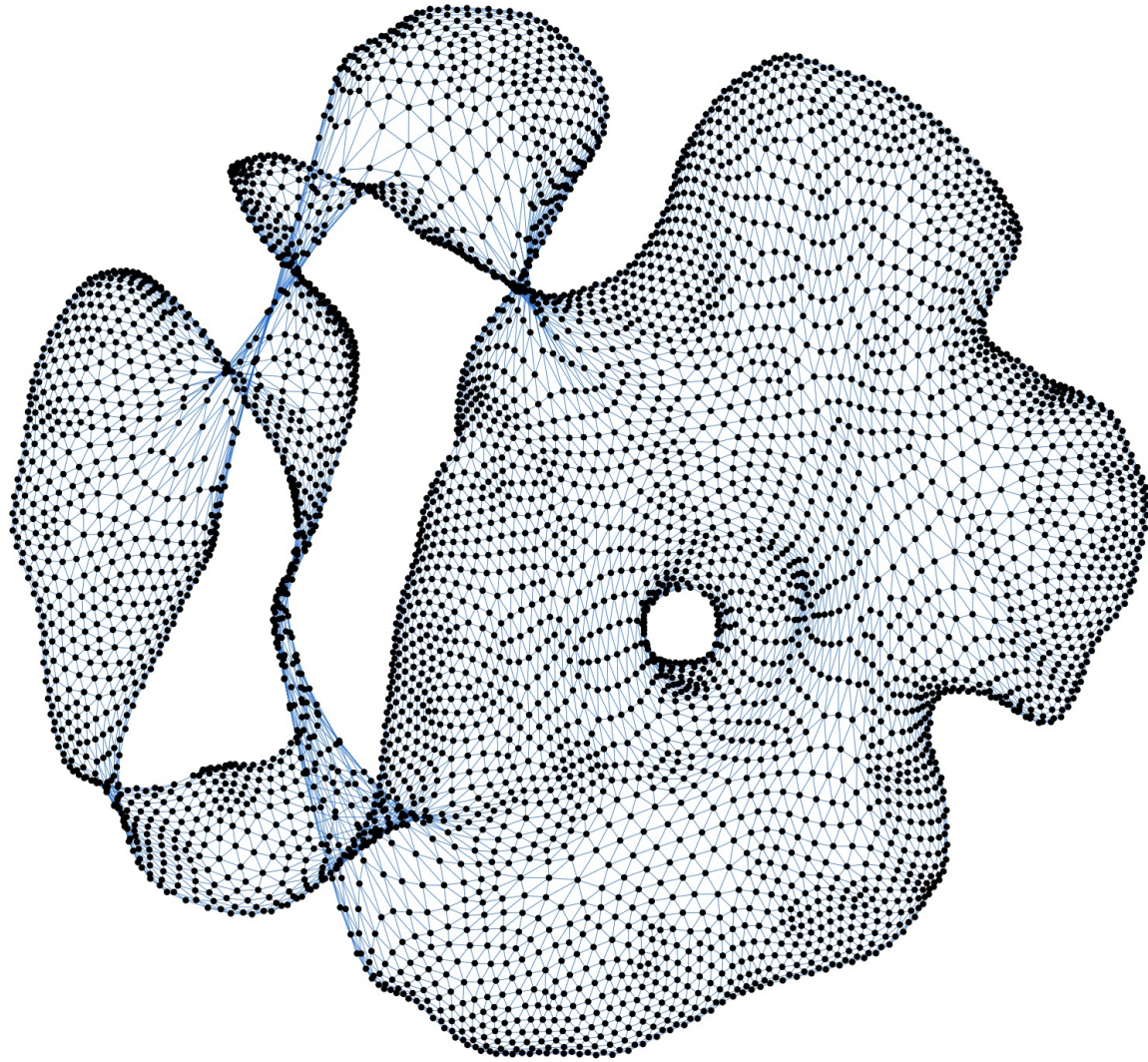
NNP-NET



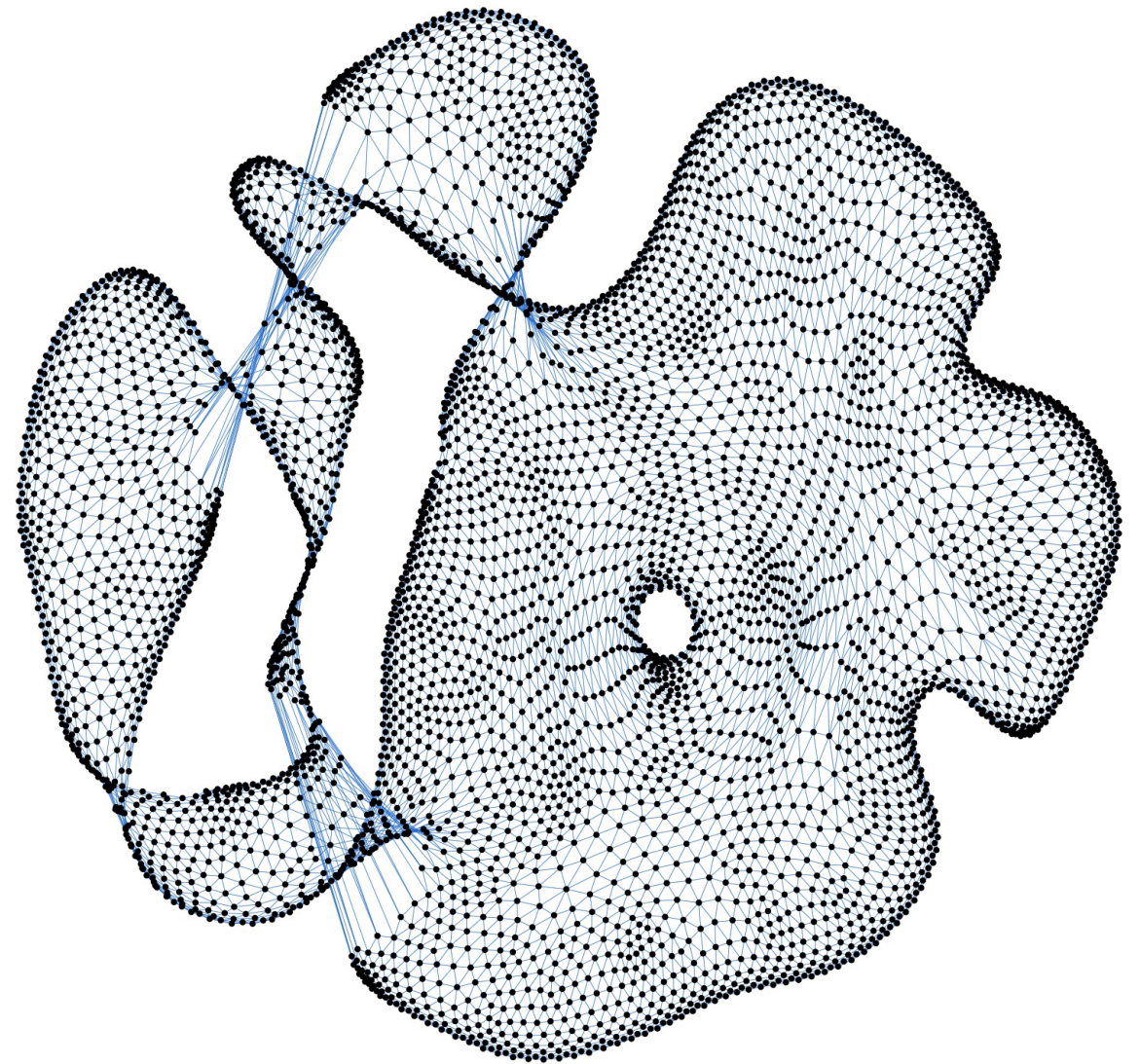
tsNET



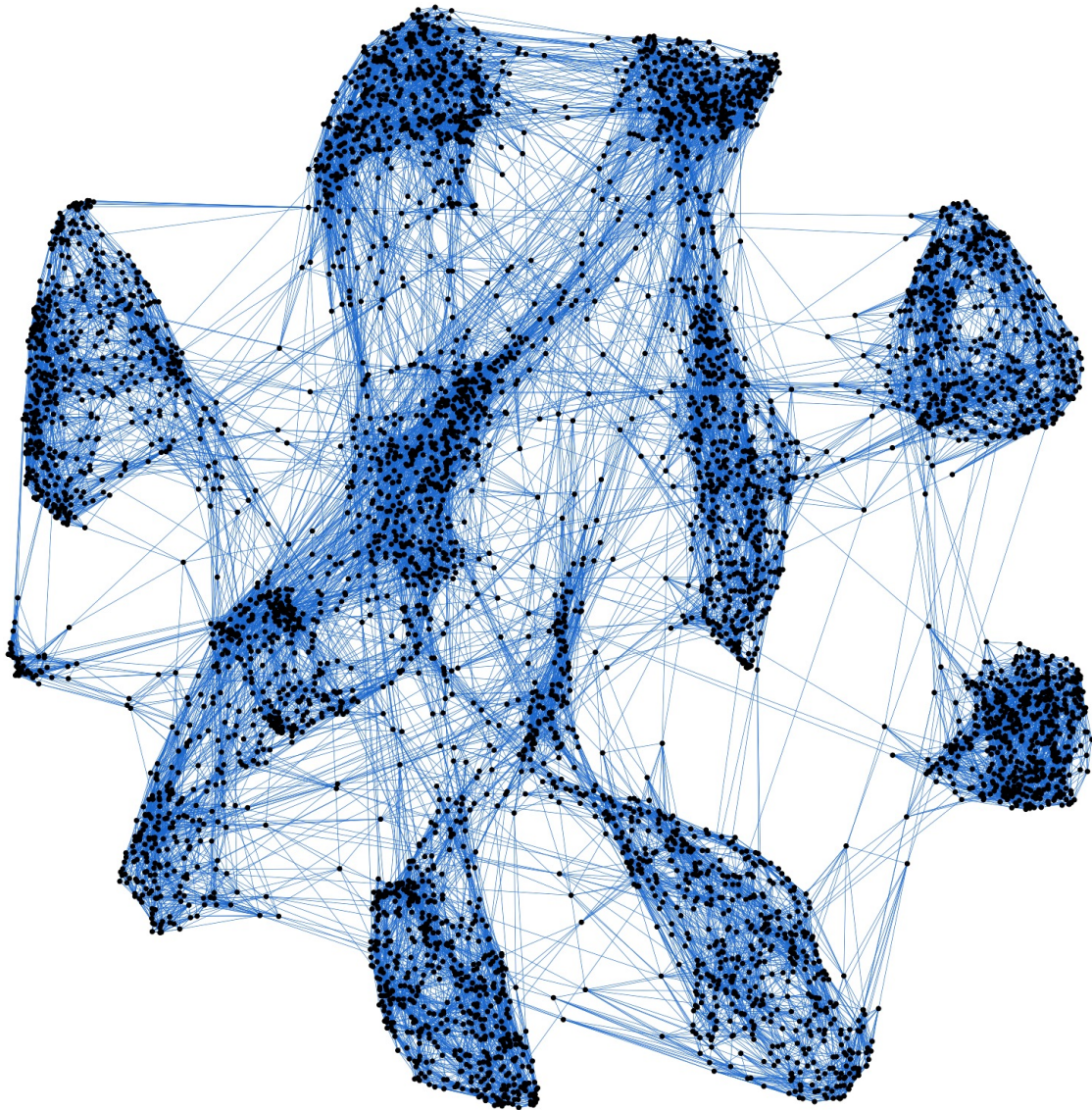
NNP-NET



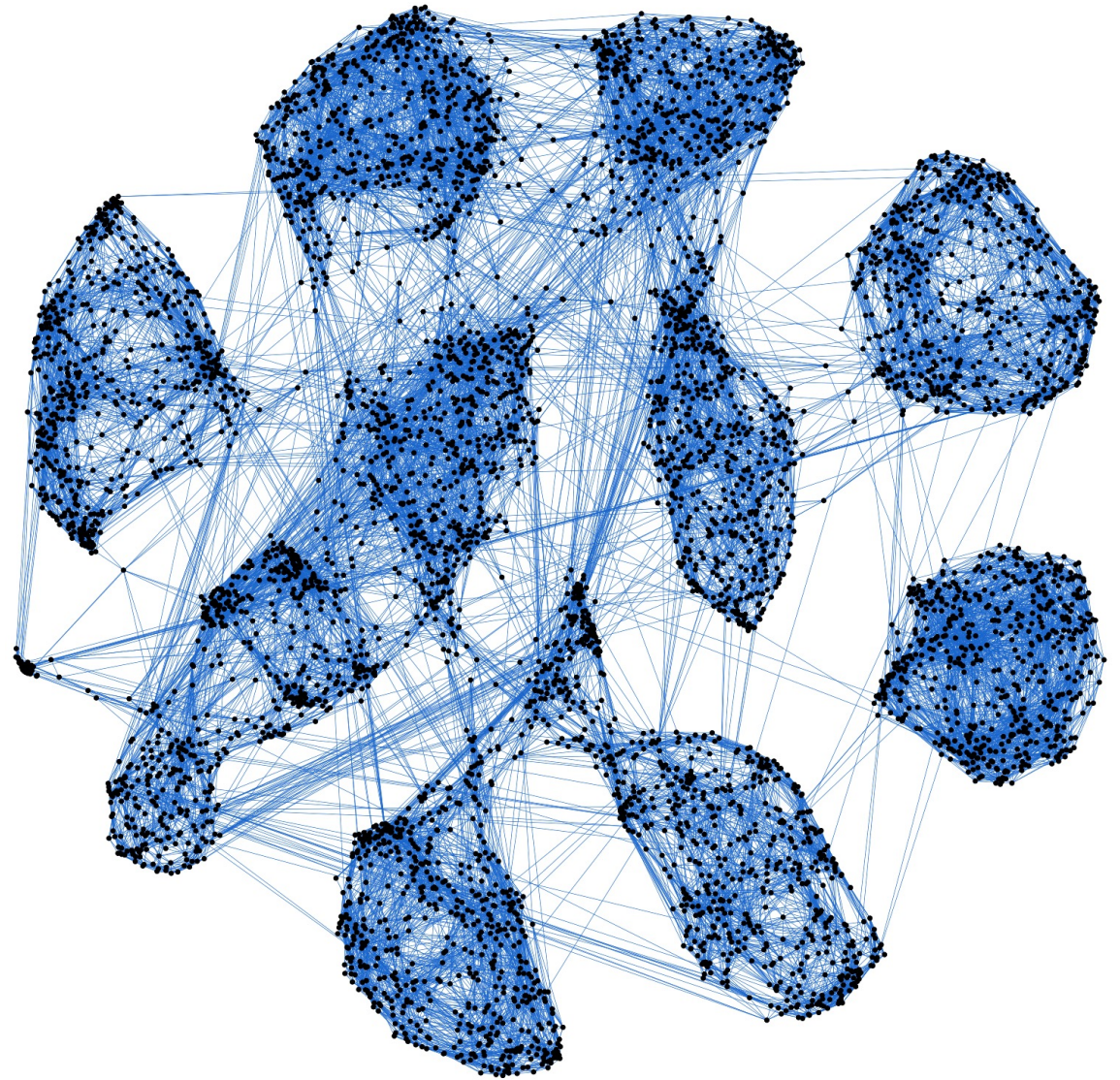
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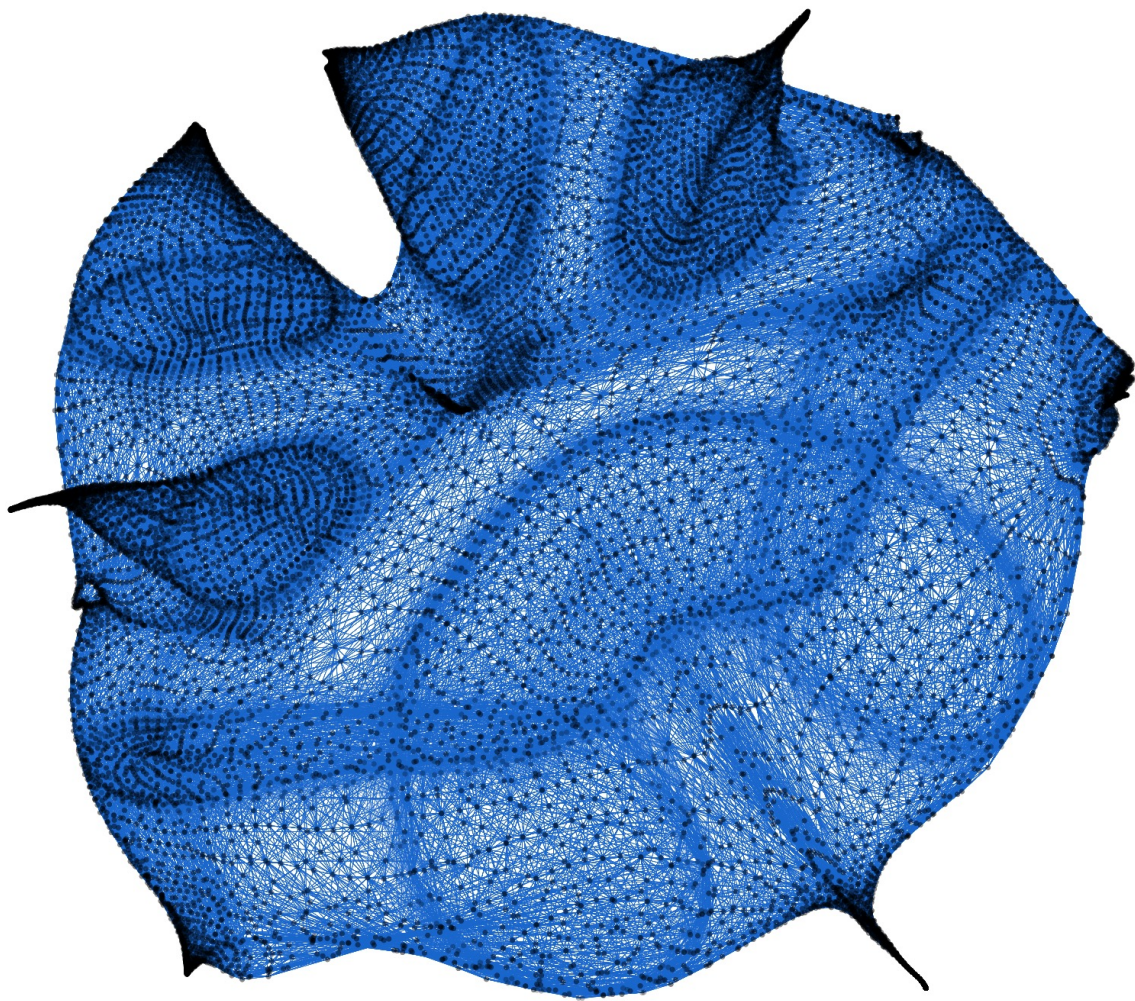
NNP-NET



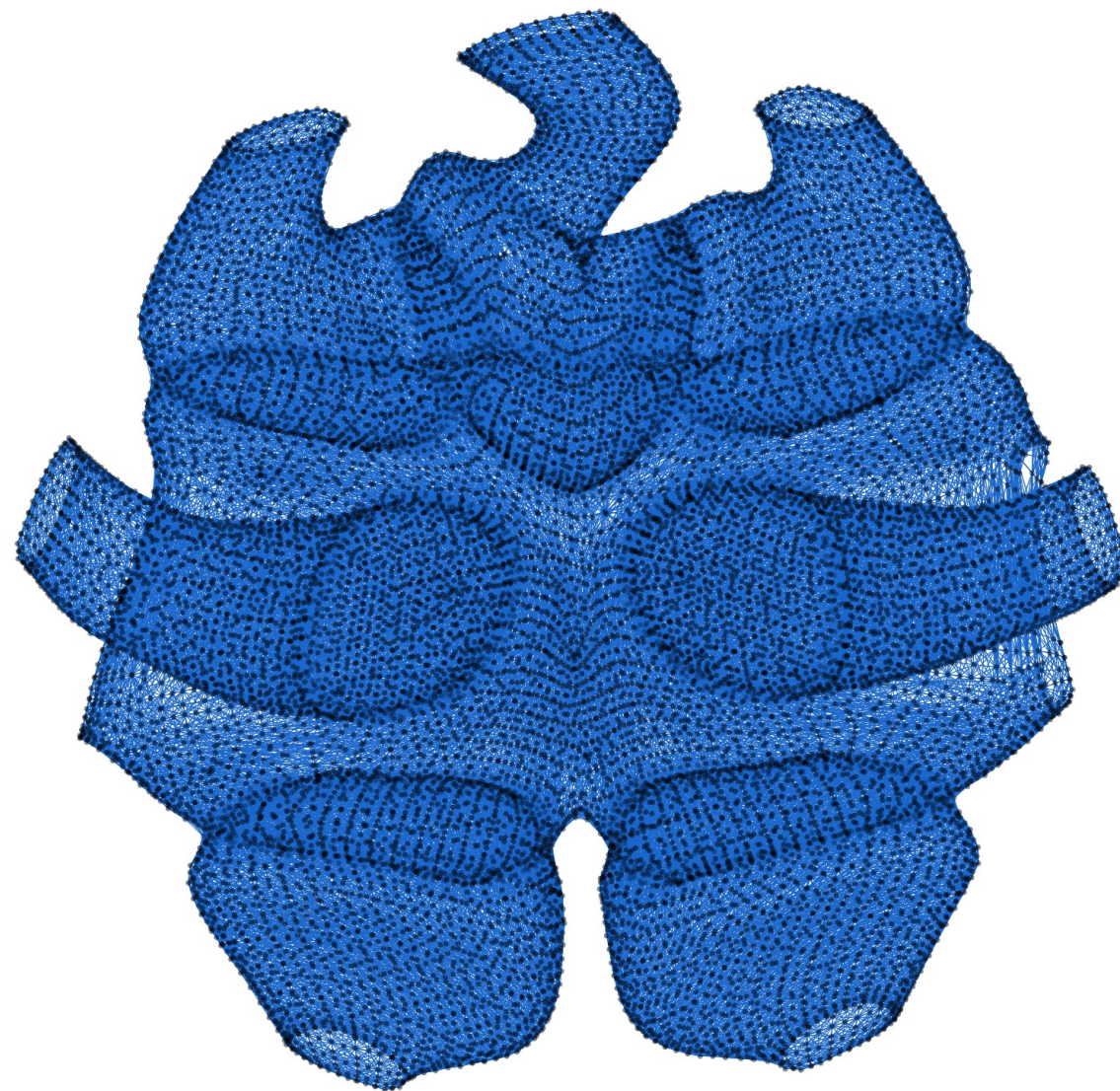
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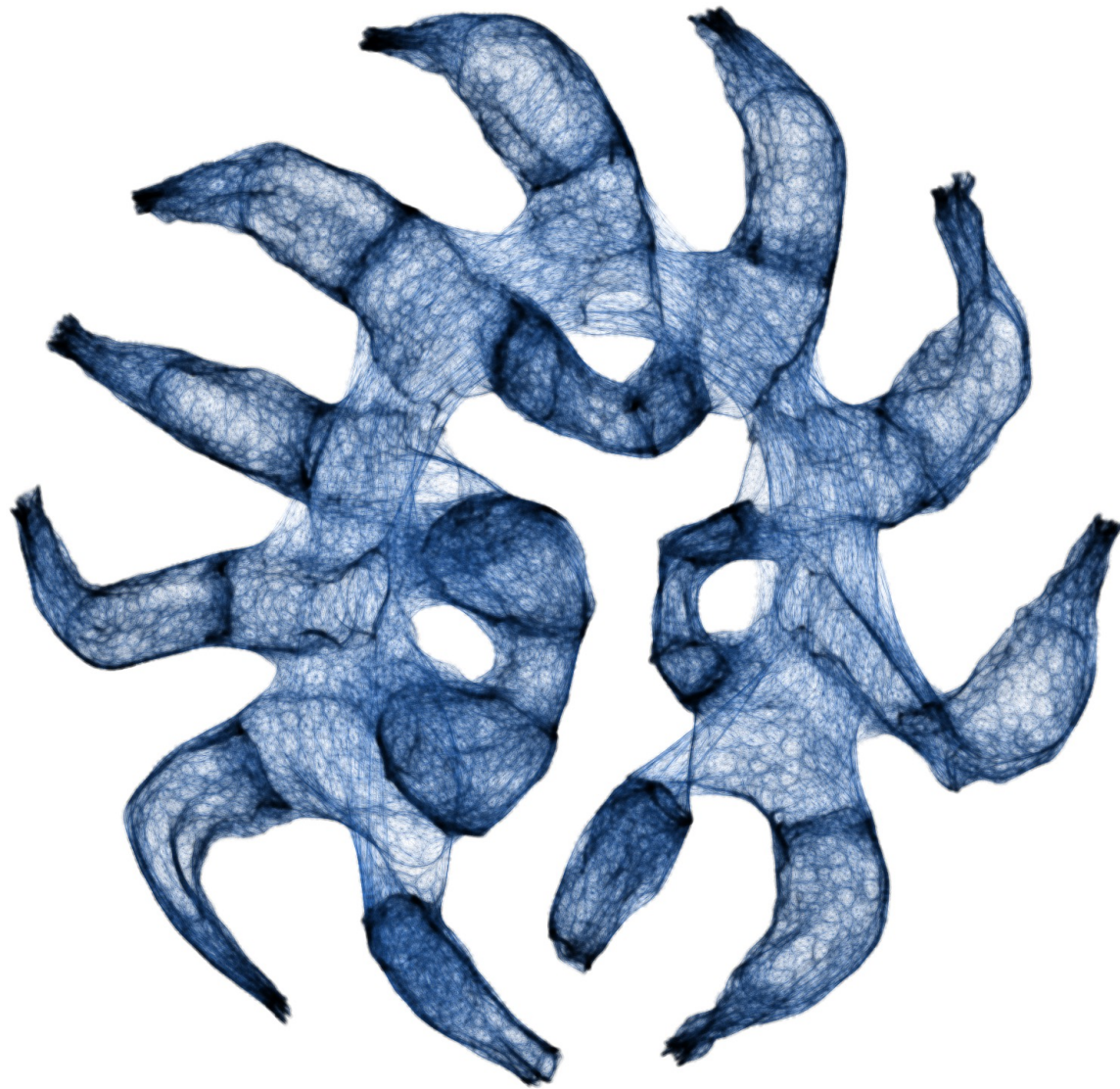
NNP-NET



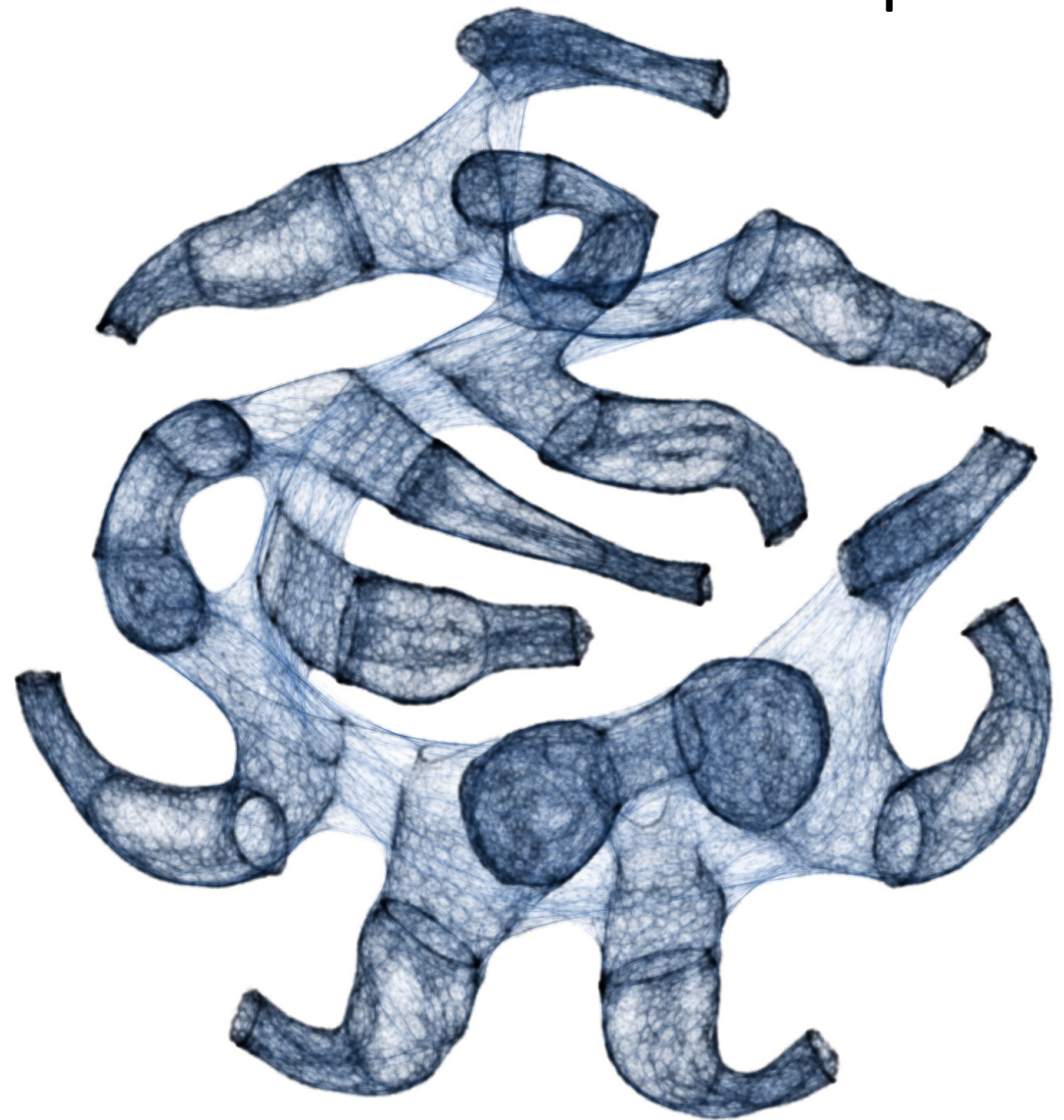
tsNET



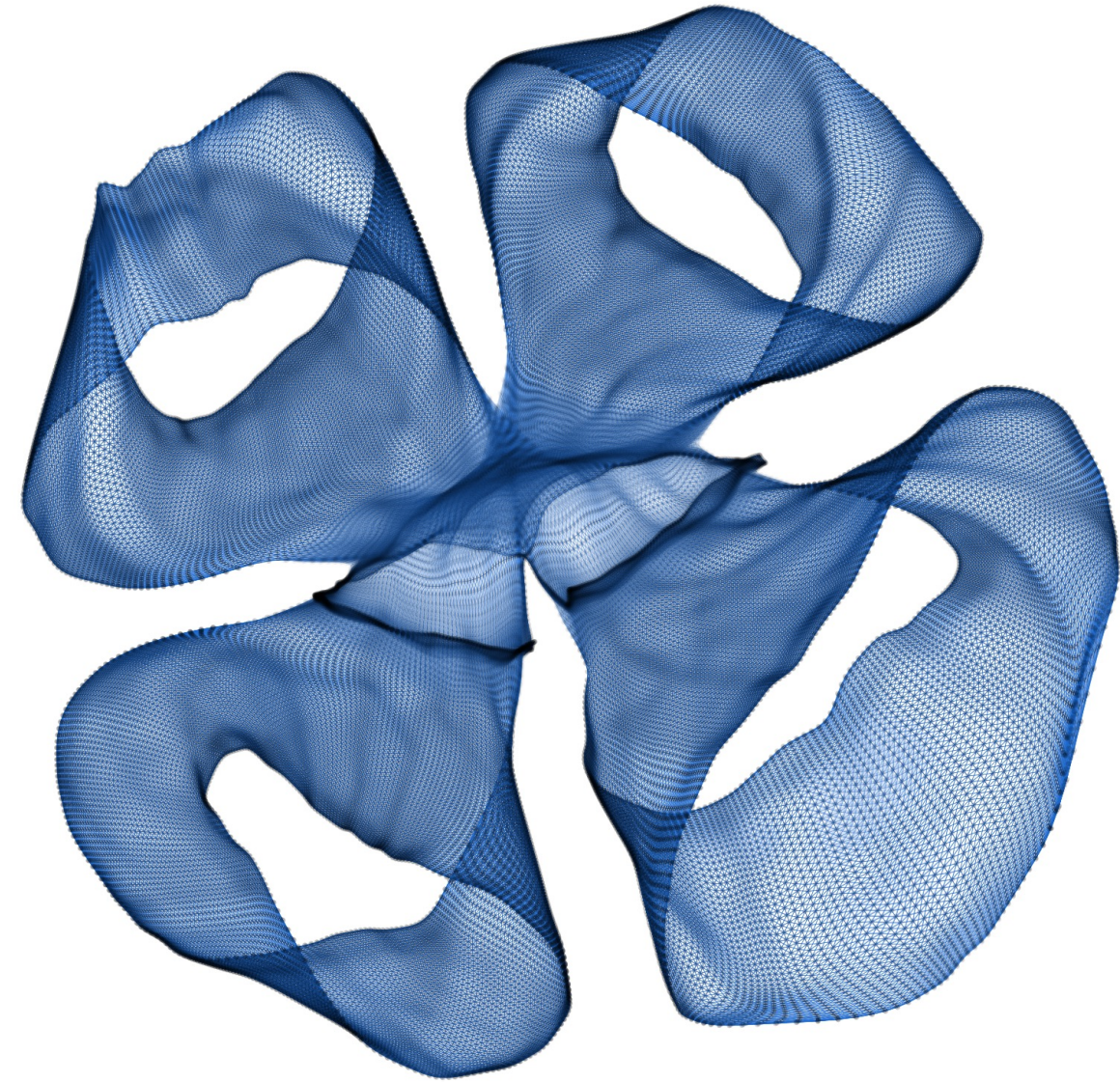
NNP-NET



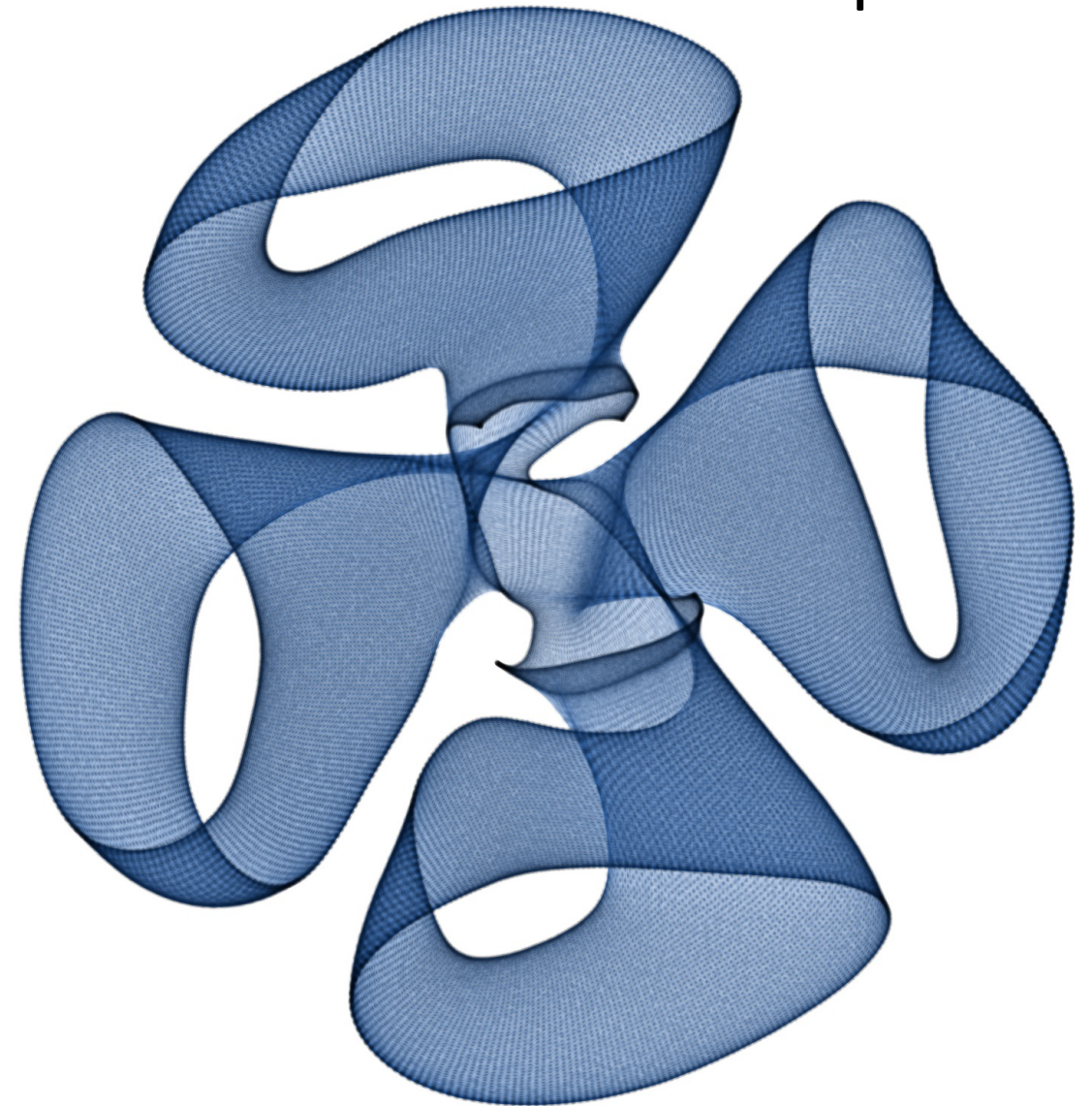
DRGraph



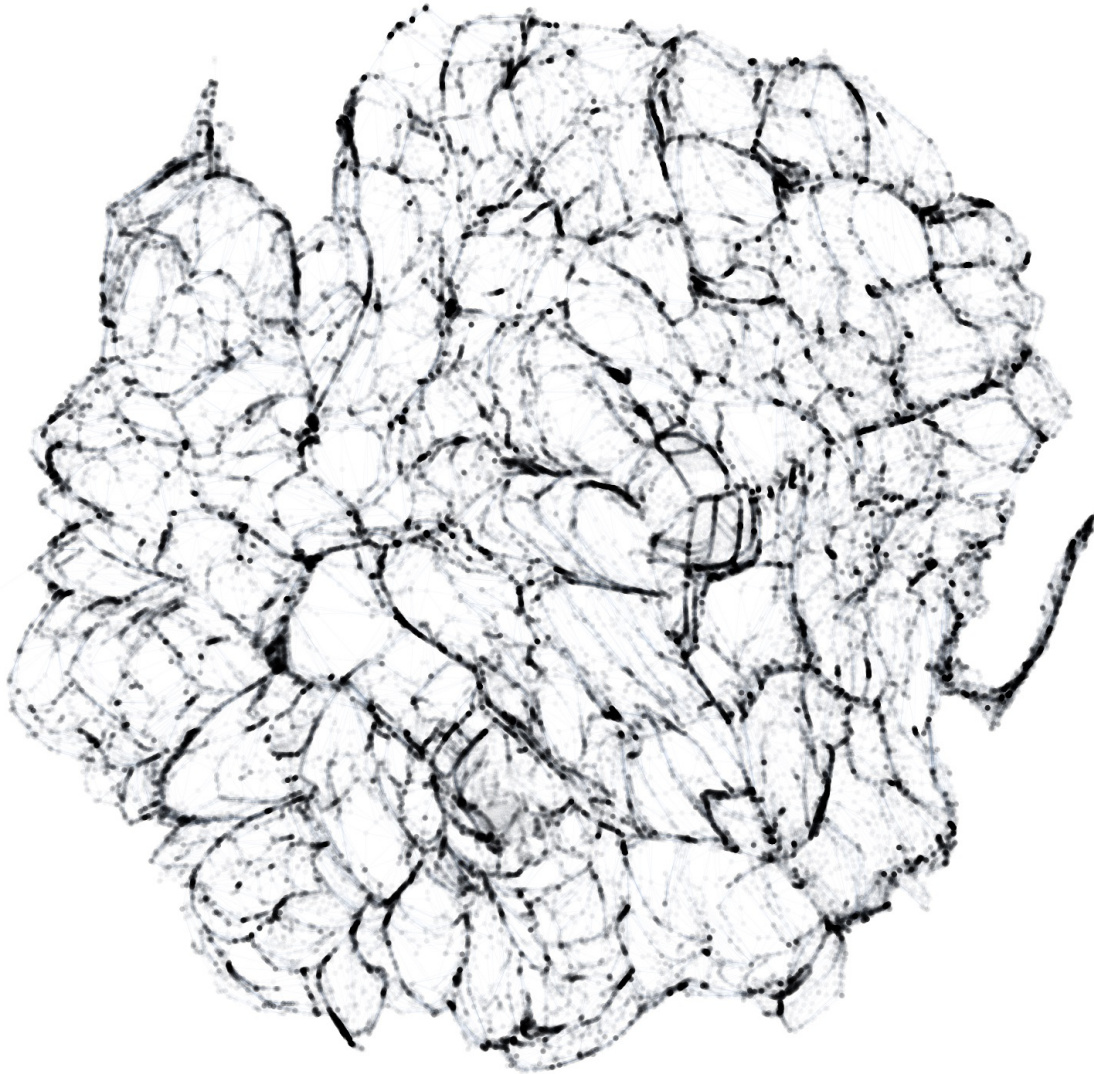
NNP-NET



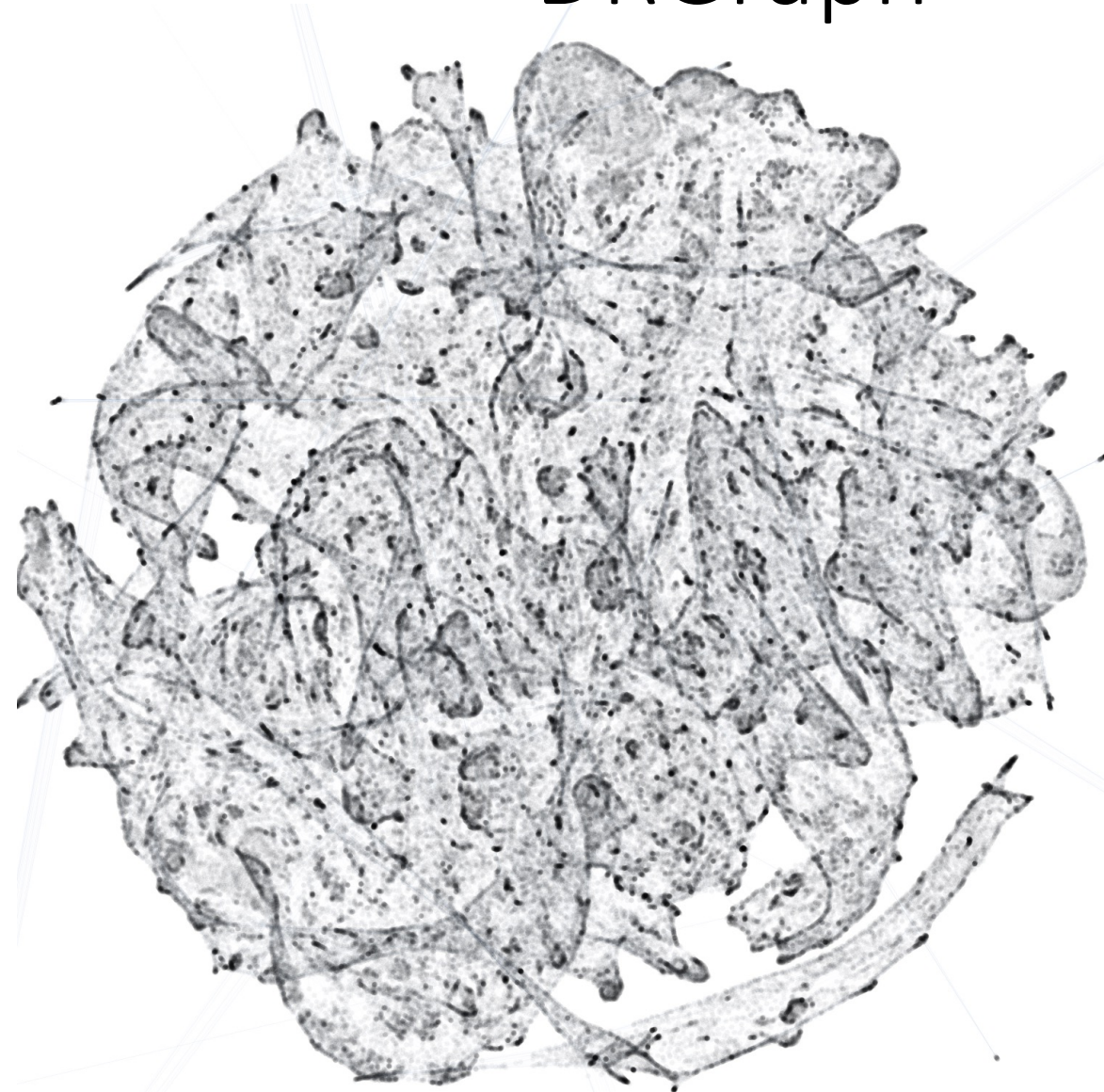
DRGraph



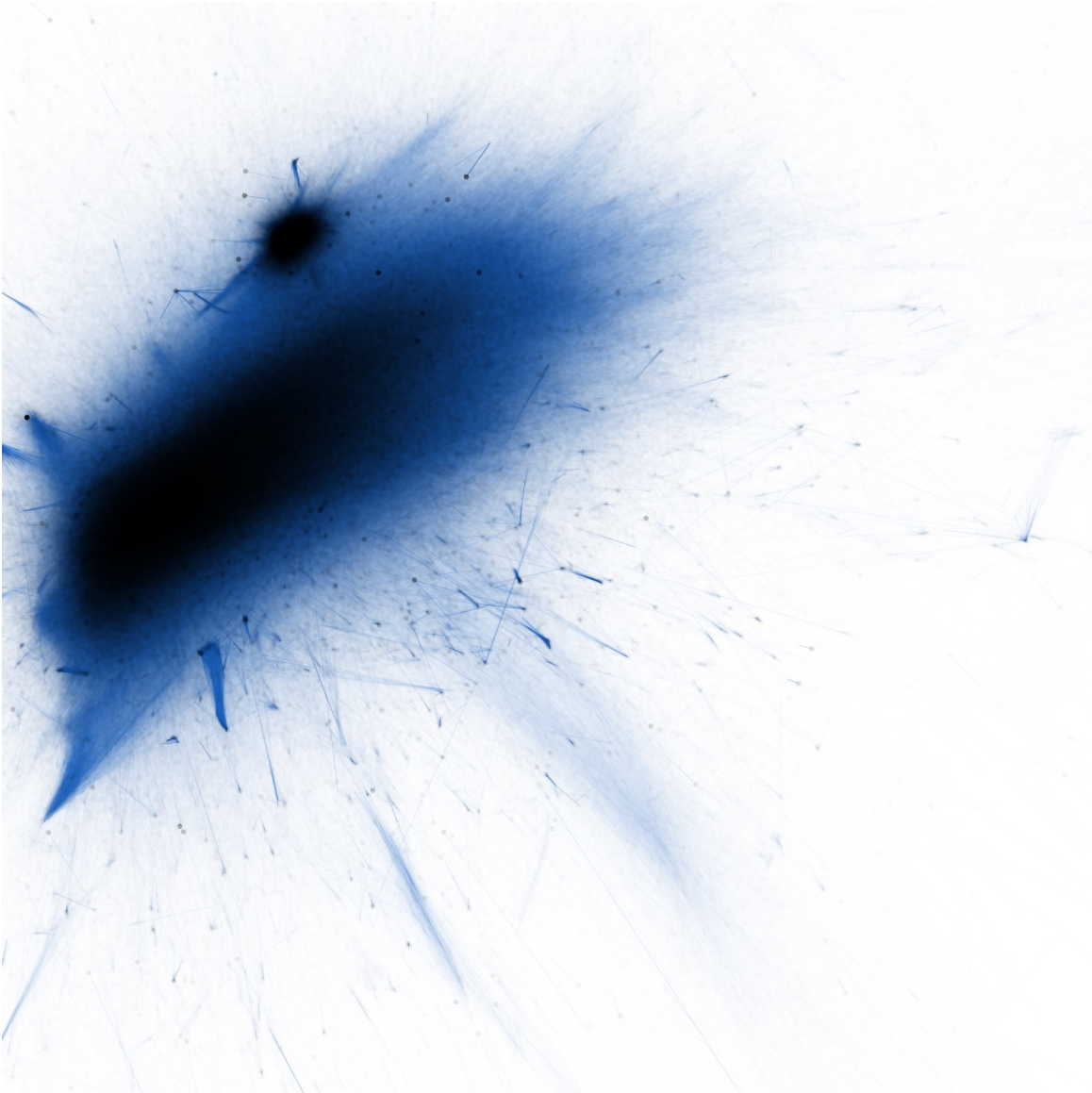
NNP-NET



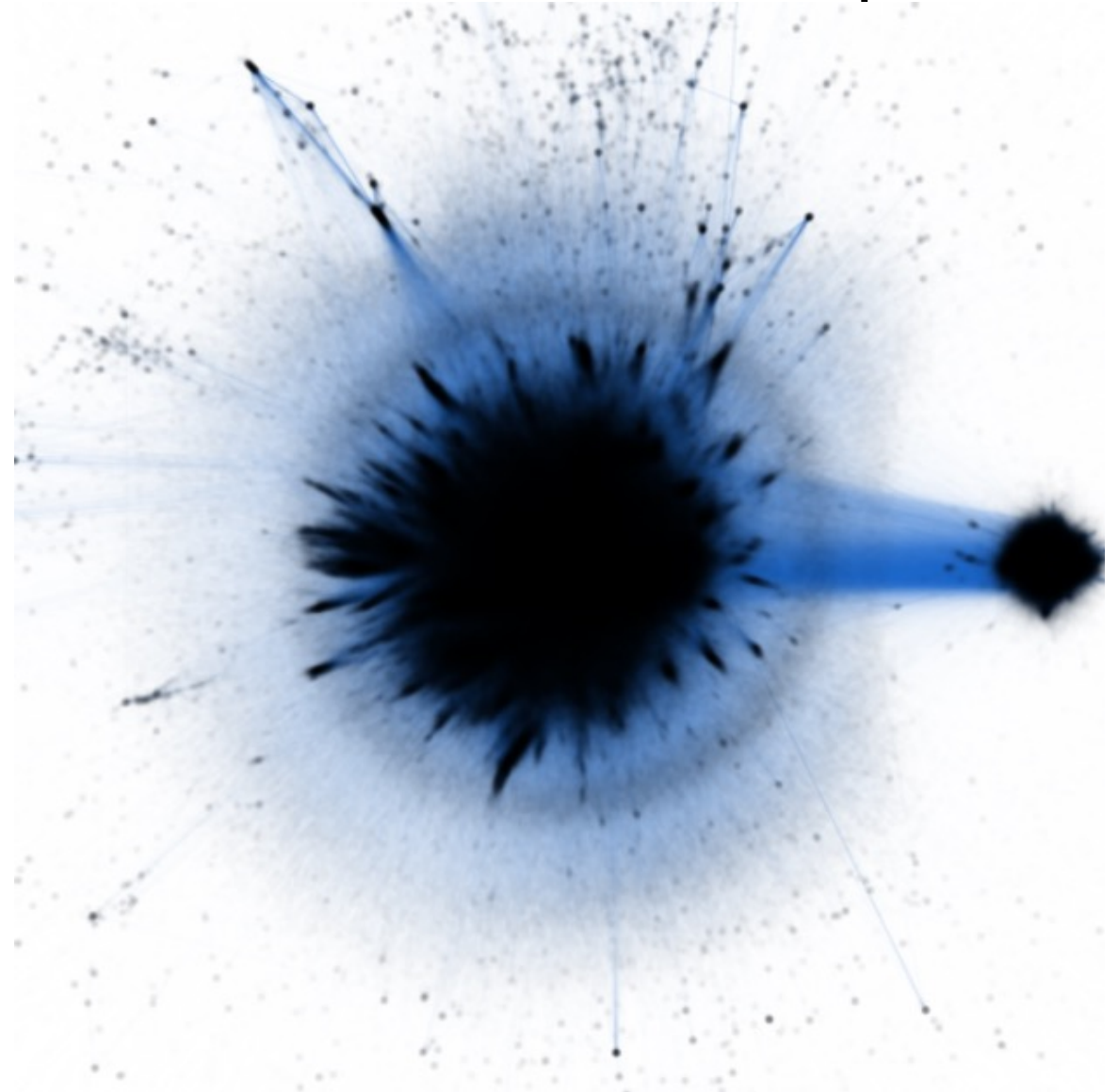
DRGraph



NNP-NET



DRGraph



Quality Metrics: Neighborhood Preservation

Number Of Nodes
↓

Graph	Neighborhood preservation per graph (higher is better)						
	SFDP	FM ³	PMDS	DRGraph	tsNET* exact	tsNET* approx	NNP-NET
dwt_1005	0.5	0.53	0.47	0.5	0.62	0.59	0.53
sierpinski3d	0.51	0.51	0.2	0.52	0.55	0.53	0.52
MISKnowledgeMap	0.17	0.16	0.13	0.33	0.4	0.41	0.24
3elt	0.62	0.65	0.36	0.63	0.66	0.63	0.63
optdigits_10NN	0.52	0.5	0.35	0.62	0.61	0.63	0.61
fe_4elt2	0.47	0.52	0.25	0.56	0.6	0.59	0.59
bcsstk36	0.45	0.41	0.3	0.49	-	0.51	0.44
k49_norm_10NN	0.048	0.045	0.034	0.12	-	0.18	0.093
fe_bcsstk32	0.32	0.38	0.21	0.41	-	0.24	0.37
m_t1	0.3	0.34	0.21	0.37	-	0.35	0.32
ship_003	0.21	0.21	0.17	0.24	-	0.2	0.24
fe_ocean	0.11	0.12	0.09	0.12	-	-	0.09
ok2010	0.48	0.46	0.27	0.32	-	-	0.44
web-NotreDame	0.38	0.31	0.33	0.46	-	-	0.39
coPapersCiteseer	-	0.079	0.058	0.17	-	-	0.091
gsm_106857	-	0.17	0.12	0.24	-	-	0.21
tx2010	0.42	0.39	0.26	0.21	-	-	0.36
com-Youtube	0.015	-	0.092	0.055	-	-	0.069
Flan_1565	-	-	0.09	0.2	-	-	0.21

Quality Metrics: Stress

Number Of Nodes
↓

Graph	Stress per graph (lower is better)						
	SFDP	FM ³	PMDS	DRGraph	tsNET* <i>exact</i>	tsNET* <i>approx</i>	NNP-NET
dwt_1005	0.029	0.026	0.029	0.03	0.038	0.036	0.029
sierpinski3d	0.079	0.078	0.104	0.08	0.077	0.086	0.125
MISKnowledgeMap	0.149	0.171	0.159	0.191	0.188	0.186	0.199
3elt	0.057	0.062	0.056	0.059	0.08	0.079	0.073
optdigits_10NN	0.159	0.144	0.125	0.172	0.142	0.14	0.139
fe_4elt2	0.05	0.069	0.064	0.068	0.095	0.098	0.097
bcsstk36	0.069	0.071	0.068	0.074	-	0.11	0.088
k49_norm_10NN	0.162	0.153	0.155	0.173	-	0.168	0.165
fe_bcsstk32	0.138	0.093	0.098	0.175	-	0.187	0.179
m_t1	0.095	0.079	0.074	0.103	-	0.078	0.147
ship_003	0.069	0.086	0.036	0.042	-	0.059	0.079
fe_ocean	0.04	0.036	0.045	0.106	-	-	0.3
ok2010	0.116	0.123	0.084	0.316	-	-	0.121
web-NotreDame	0.226	0.178	0.319	0.492	-	-	0.314
coPapersCiteseer	-	0.185	0.249	0.245	-	-	0.215
gsm_106857	-	0.089	0.102	0.131	-	-	0.106
tx2010	0.13	0.159	0.058	0.705	-	-	0.137
com-Youtube	0.179	-	0.275	0.367	-	-	0.427
Flan_1565	-	-	0.093	0.093	-	-	0.089
com-LiveJournal	-	-	0.206	0.305	-	-	0.27

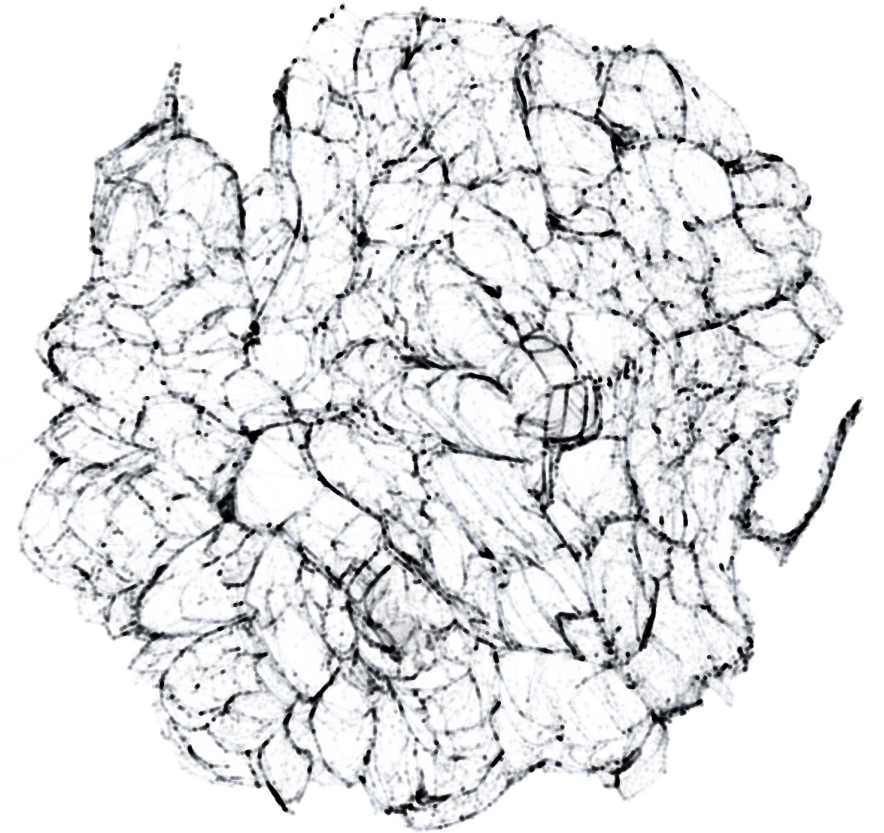
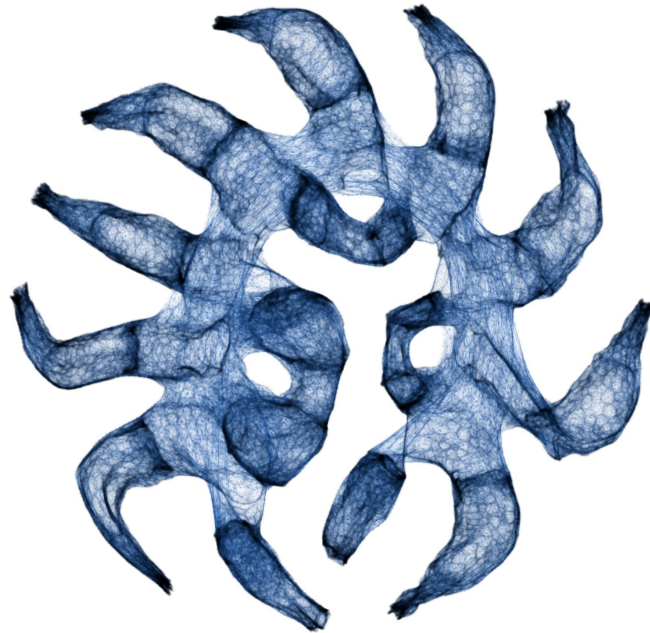
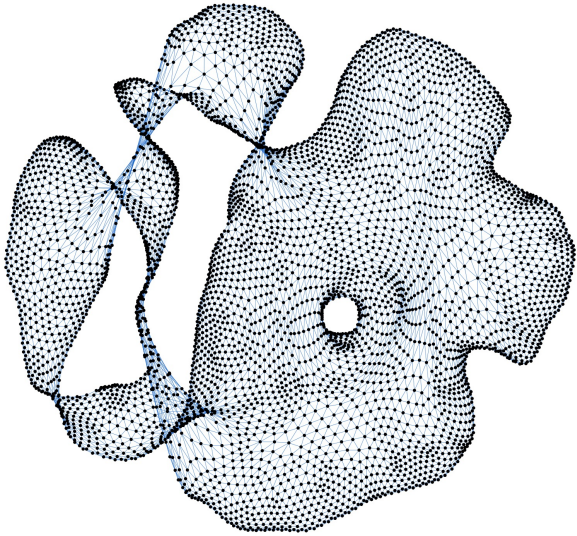
Future work

- Improve the subgraph method
- Other ground truth method
- Embedding method

Conclusion

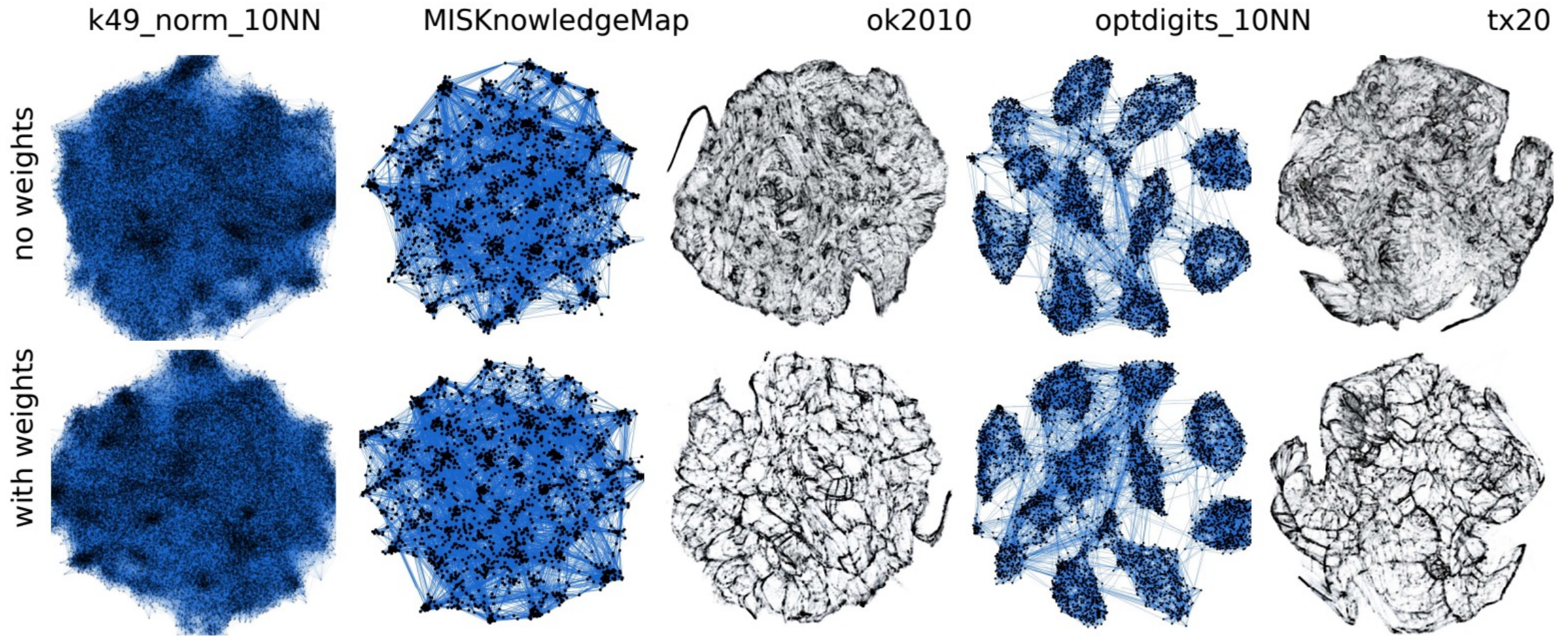
- Competitive results on graphs > 1 million nodes
 - Better execution time than DRGraph
 - Similar quality metrics
- ML-based DR method successfully used for GD
- GD and DR are very related

Questions?

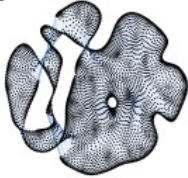
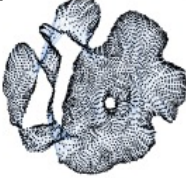
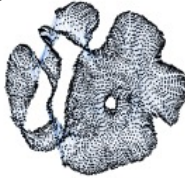
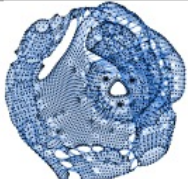
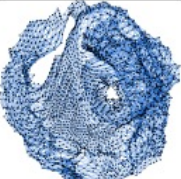
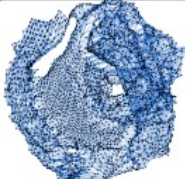
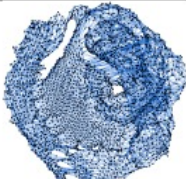
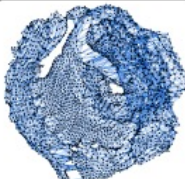
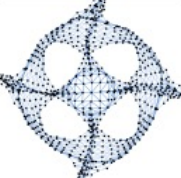
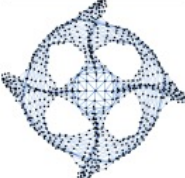
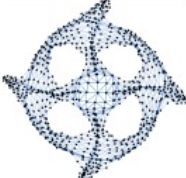
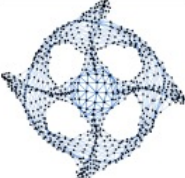
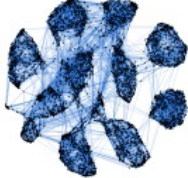
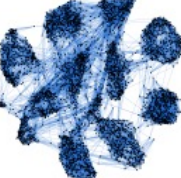
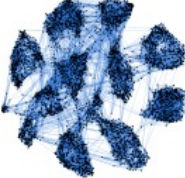
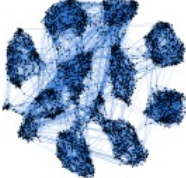
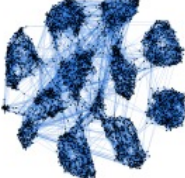
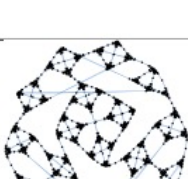
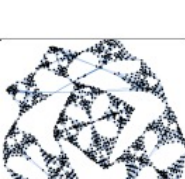
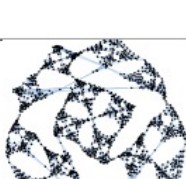


Source code can be found at <https://github.com/IlanHartskeerl/NNP-NET>

Weight Comparison



Subgraph size

Graph	Ground Truth	$n = 10$	$n = 25$	$n = 50$	$n = 100$
3elt					
bcsstk36					
dwt_1005					
optdigits_10NN					
sierpinski3d					

All execution time comparison

Graph	Execution time per method						
	SFDP	FM ³	PMDS	DRGraph	tsNET* <i>exact</i>	tsNET* <i>approx</i>	NNP-NET
dwt_1005	0.34	0.12	0.02	0.06	1.44	2.61	10.13
sierpinski3d	0.75	0.12	1.37	0.11	6.52	5.42	11.38
MISKnowledgeMap	1.23	0.76	1.20	0.13	40.42	13.71	24.27
3elt	1.71	0.20	0.07	0.27	110.70	18.92	21.14
optdigits_10NN	2.59	0.58	2.38	0.30	155.46	59.68	36.88
fe_4elt2	6.06	0.22	0.15	0.59	931.18	107.97	58.96
bcsstk36	10.89	2.28	0.53	1.49	-	288.91	56.96
k49_norm_10NN	37.28	2.14	37.13	2.66	-	1718.79	156.79
fe_bcsstk32	28.25	3.07	2.47	3.64	-	572.96	86.68
m_t1	77.61	14.22	4.04	8.21	-	1133.24	87.81
ship_003	94.44	14.24	5.85	9.61	-	1456.06	66.40
fe_ocean	117.74	1.80	2.28	14.39	-	-	94.24
ok2010	260.64	4.13	44.83	23.98	-	-	94.18
web-NotreDame	270.48	10.26	6.13	25.49	-	-	166.56
coPapersCiteseer	-	99.34	23.69	45.49	-	-	98.20
gsm_106857	-	57.28	35.35	55.86	-	-	160.41
tx2010	1165.55	12.95	200.21	83.18	-	-	254.17
com-Youtube	2209.56	-	23.47	132.47	-	-	1293.10
Flan_1565	-	-	55.92	171.44	-	-	124.12
com-LiveJournal	-	-	220.14	618.30	-	-	394.85